

PR200 OPERATOR TRAINING

PR200 APPS

ROHDE & SCHWARZ

Make ideas real





PR200 APPS – OVERVIEW

PR200 ACCESSING APPS

Starting an app

To access the apps, press the [App] key.

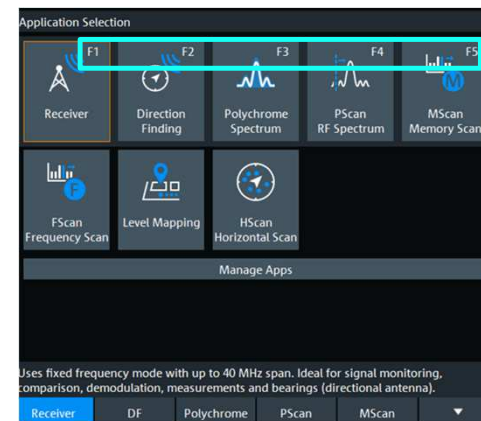
The "Application Selection" dialog appears in the display.

The softkeys show which apps are assigned to them.

To start an app, use any of the following actions:

Select the app in the "Application Selection" dialog (e.g. using the front rotary control), then activate it (e.g. using the [Enter] key or the [Ok] button in the center of the rotary control).

Press the softkey to which the app has been assigned.



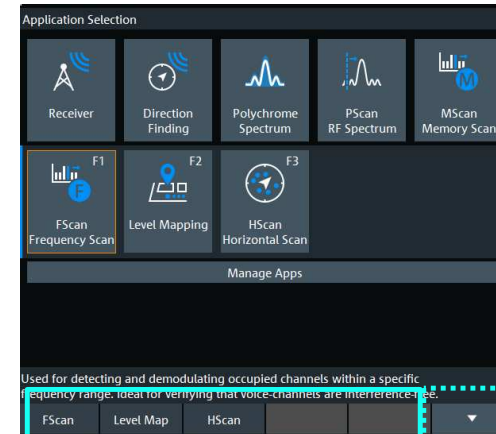
PR200 ACCESSING APPS

Softkeys assignment to next row

Assigning softkeys to the next row, press the “F6” key.

Softkeys will be assigned to the next row.

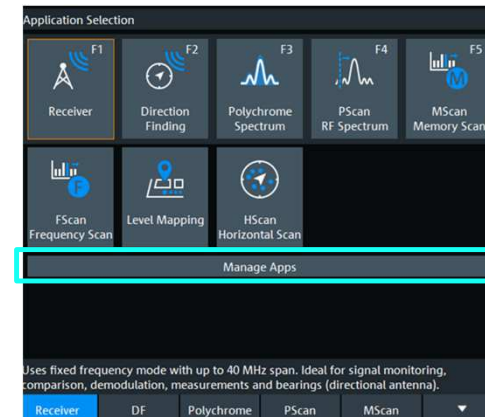
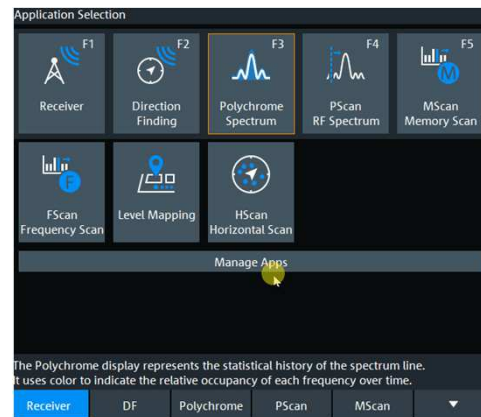
Softkeys’ names will change to corresponding APPs’ names belonging in the next row



PR200 ACCESSING APPS

Moving the icons

To move the icons, select “Manage Apps” by means of the rotary control and the “Enter” or “OK” button.



PR200 APPS

Available Apps

- Receiver (i.e. FFM)
- Polychrome Spectrum (option CS-PC)
- PSCAN (option CS-PS)
- Level mapping (option CS-MAP)
- MScan
- HScan (option CS-MAP)
- FScan



PR200 APPS – RECEIVER

PR200 RECEIVER

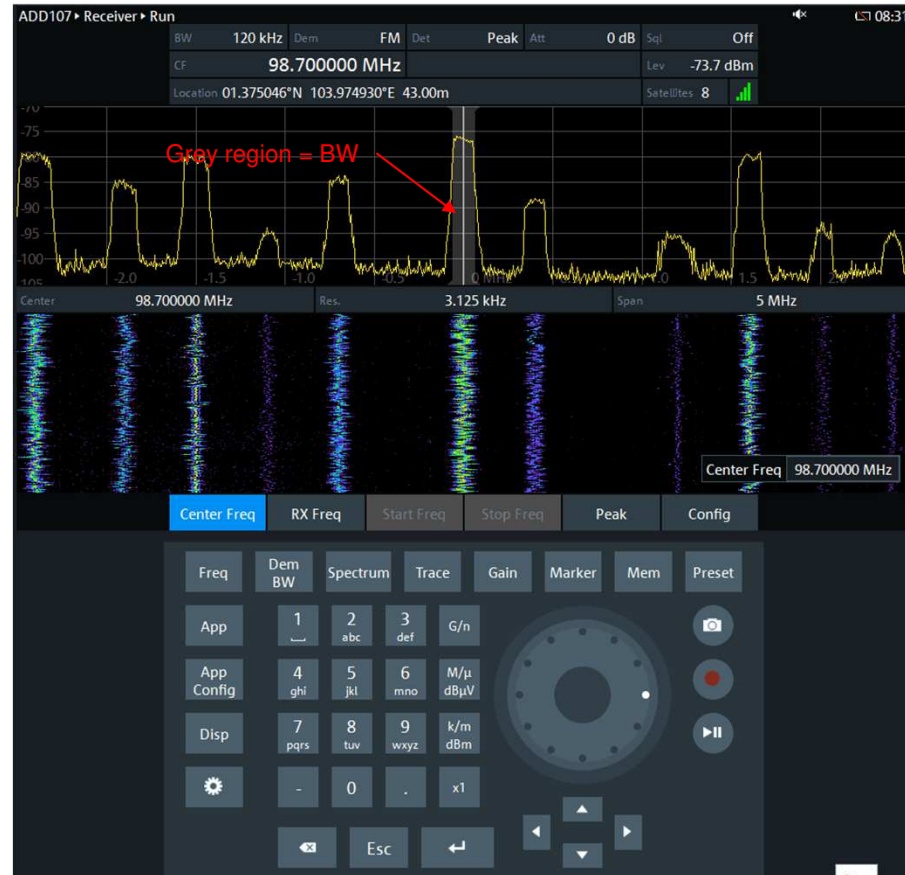
APP

–Receiver (i.e. FFM)

This app puts the R&S PR200 into fixed frequency mode (FFM).

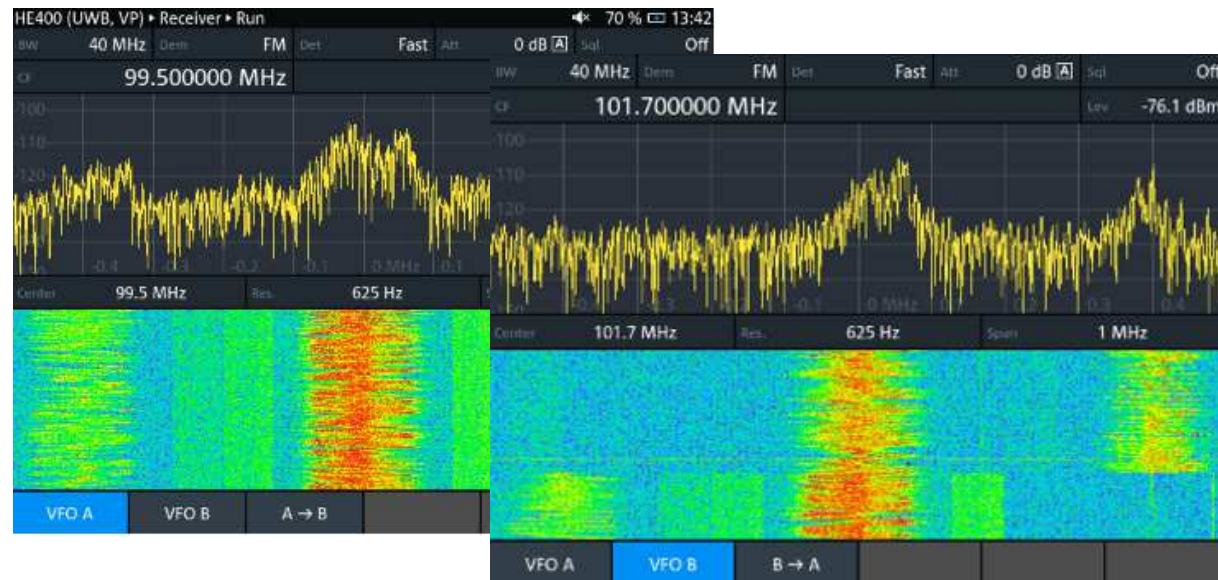
Possible operations:

- ▶ Continuous monitoring of a particular scenario (up to 40 MHz wide)
- ▶ Demodulation of a particular signal
- ▶ Comparison/surveillance of two signals ("VFO A" and "VFO B")
- ▶ Record/replay a particular signal/scenario
- ▶ Determine the bearing to a particular transmitter (directional antenna)



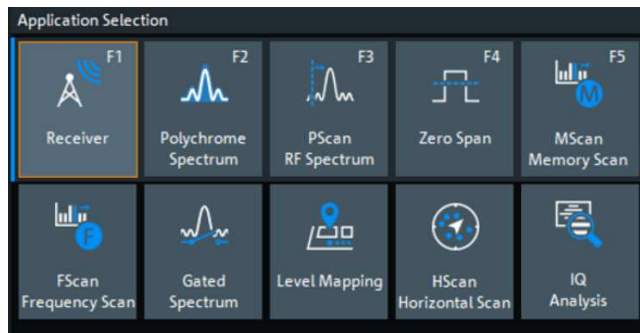
DUAL VARIABLE FREQUENCY OSCILLATOR (VFO)

- ▶ Can be accessed using the App Config button while in the Receiver App
- ▶ In Receiver mode you can rapidly toggle between two different frequencies (VFO-A and VFO-B). Each VFO can have its own parameters (Span, Demod BW/Type, etc)
- ▶ Useful for looking at two signals of interest in quick succession
 - Uplink/downlink

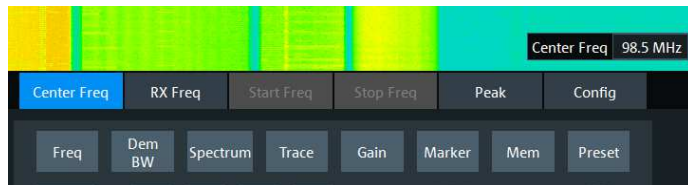


PR200 RECEIVER

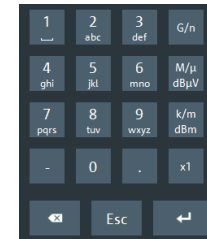
1. Select “Receiver” using the arrow keys or rotary control and press the center of the rotary control or the “Enter” key.



2. Press “Freq” on the top row and follow by the “Center Freq” (F1).



3. Enter the desired frequency via the number pad and press the “Enter” key or the appropriate units key to accept the entry.



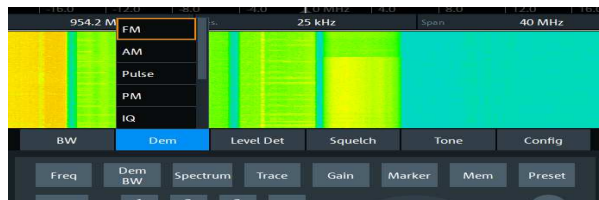
4. Press the “Dem BW” and follow by “BW” (F1).



5. Select the desired bandwidth by the rotary control or the arrow keys.

PR200 RECEIVER

- Press “Dem” (F2) and select the desired demodulation by the rotary control or arrow keys.



- Press “Level Det” (F3) follow by select field for “Level 1” and select the desired detector. 3 level bars can be enabled.



- Press “Squelch” (F4) and toggle it on or off when needed. Threshold level can be adjusted here.



- Press “Config” (F5) follow by select “Meas mode” to change the desired measurement mode.

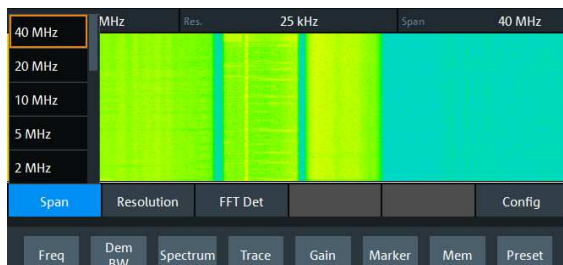


PR200 RECEIVER

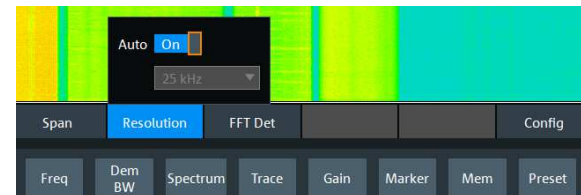
10. Select “Meas time” to change the measurement time mode. In manual mode, desired measurement time can be specified.



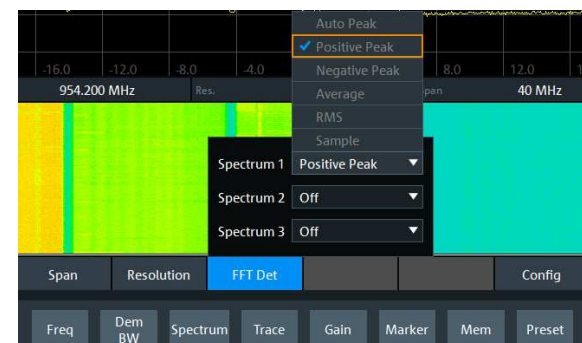
11. Press “Spectrum” follow by select “Span” (F1) to set the lower and upper frequency range of the spectrum display ($\pm \text{Span} / 2$).



12. Select “Resolution” (F2) and set the desired frequency step. The smaller the value, the higher the resolution. Default is Auto On. Toggle Off to set the step size manually.

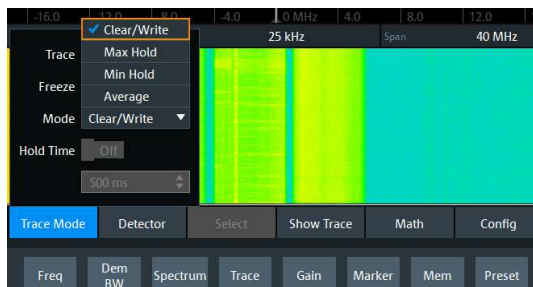


13. Select “FFT Det” (F3) follow by “Spectrum 1” to set the desired detector mode. 3 spectrum traces can be enabled.

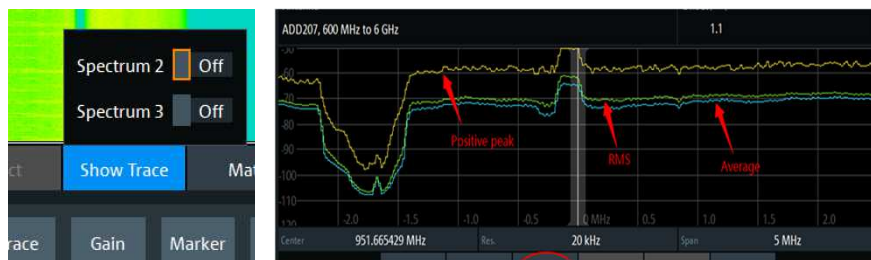


PR200 RECEIVER

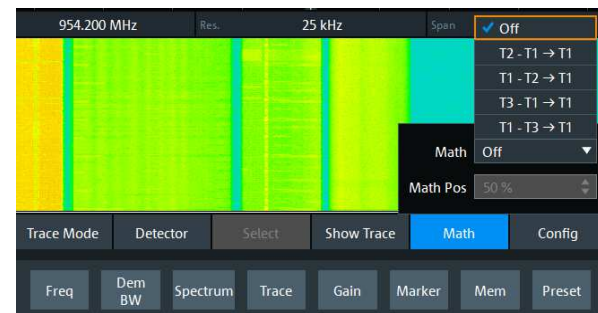
- Press “Trace” follow by “Trace Mode” (F1) to change the “Mode” of the trace display.



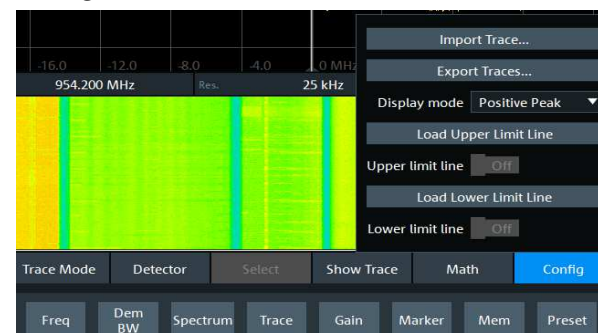
- Press “Show Trace” (F4) and toggle on additional spectrums to be displayed.



- Press “Math” (F5) to enable arithmetic operation between spectrum traces.



- Press “Config” (F6) to import or export traces and to enable limit lines. Limit line creation is performed using R&S InstrumentView software tool.



PR200 RECEIVER

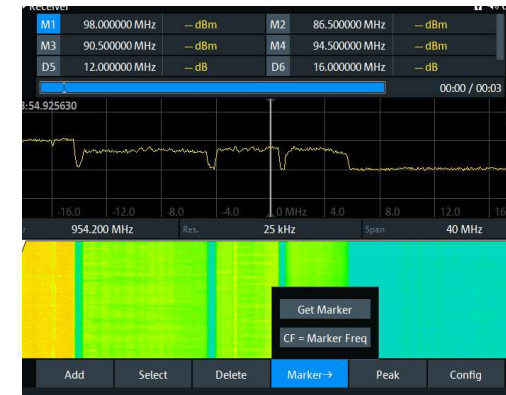
18. Press “Marker” to add markers. 4 types of markers can be added : Spectrum marker, spectrum line, waterfall line and delta marker.

Maximum number of spectrum marker depends on the number of delta marker added, i.e. 4 spectrum markers and 2 delta markers, 6 spectrum markers and 0 delta marker.

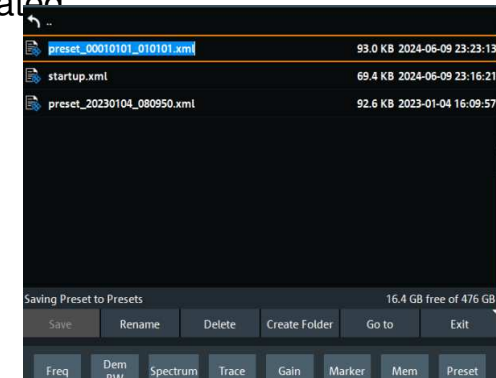
Maximum of 2 lines each for spectrum and waterfall lines.



19. To bring a selected marker which is out of frequency range back into the current range, press “Marker->” (F4) and select “Get Marker”.

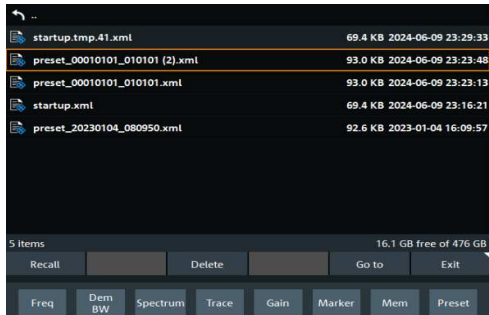


20. Press “Preset” follow by “Save” (F1) to open the directory window. Press “Save” again and a preset file will be created



PR200 RECEIVER

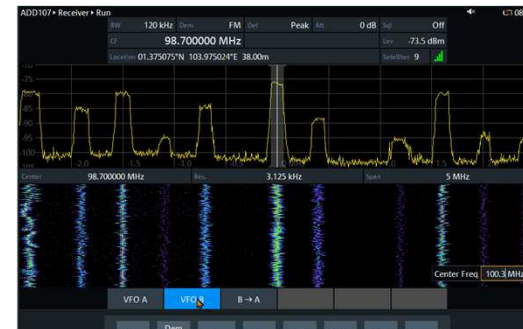
21. Press “Preset” follow by “Recall” (F1) to open the directory window. Select the preset file and press “Recall”.



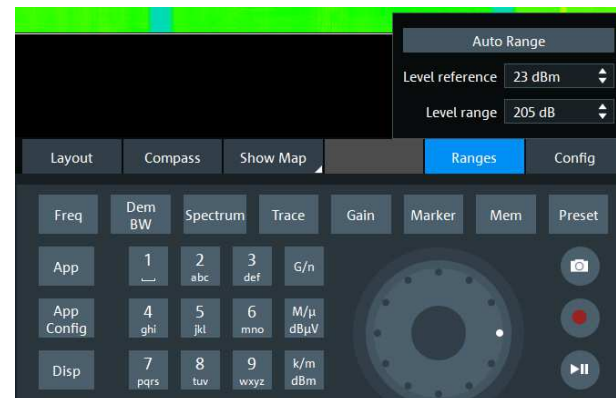
22. To use VFOs, press “App Config”. The current frequency is already on “VFO A” (F1).



23. Press “VFO B” (F2) and press the number pad to enter the frequency to be assigned to “VFO B”.

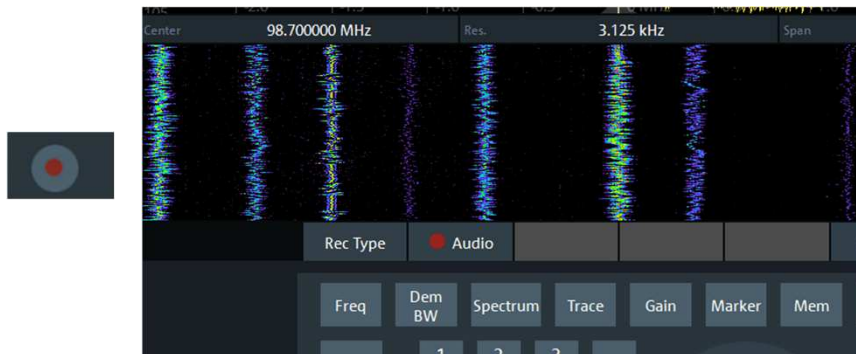


24. To autorange the spectrum display, press the “Disp” button and then the “Ranges” (F5) button.

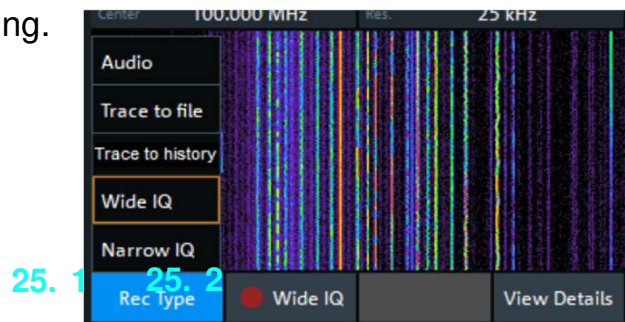


PR200 RECEIVER

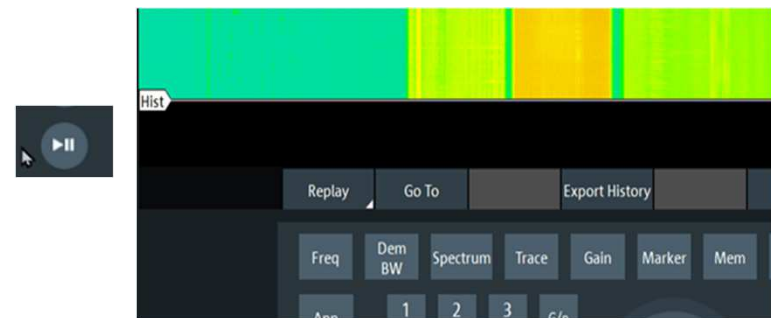
25. To perform recording, press the record button to open the recording dialog.



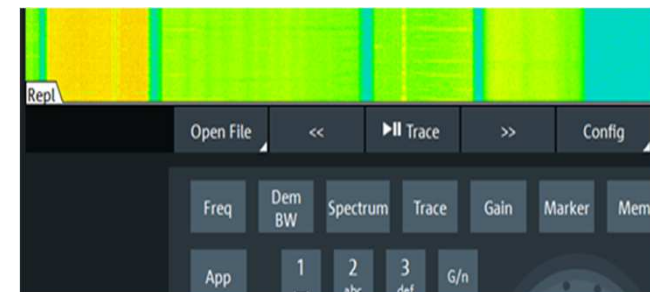
26. Press “Rec Type” (F1) to select the desired content to record then press the record button to start the recording. Press the record button again to stop recording.



27. To replay a recording, press the “Play/Pause” button to open the “Replay” function and press “Replay” (F1).

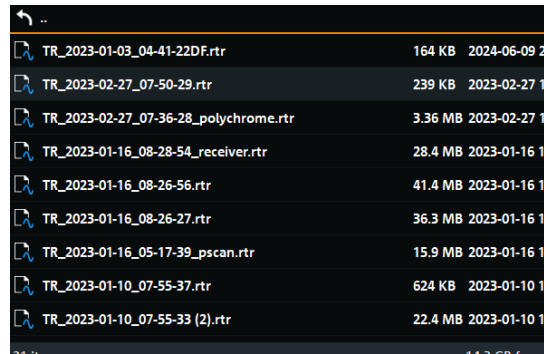


28. Press “Open File” (F1) to open file list.

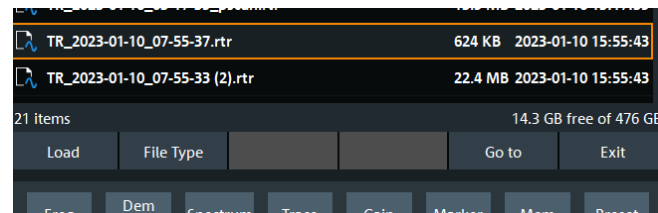


PR200 RECEIVER

29. Select the file to load by using the rotary control.



30. Press “Load” (F1) to load the selected file for replay.



30. Press the “Play/Pause” button to start playing.



HANDS ON

- ▶ Receiver Touch & Feel
- ▶ Play FM Radio
- ▶ Record a short audio
- ▶ *(est. 15 - 30 min)*



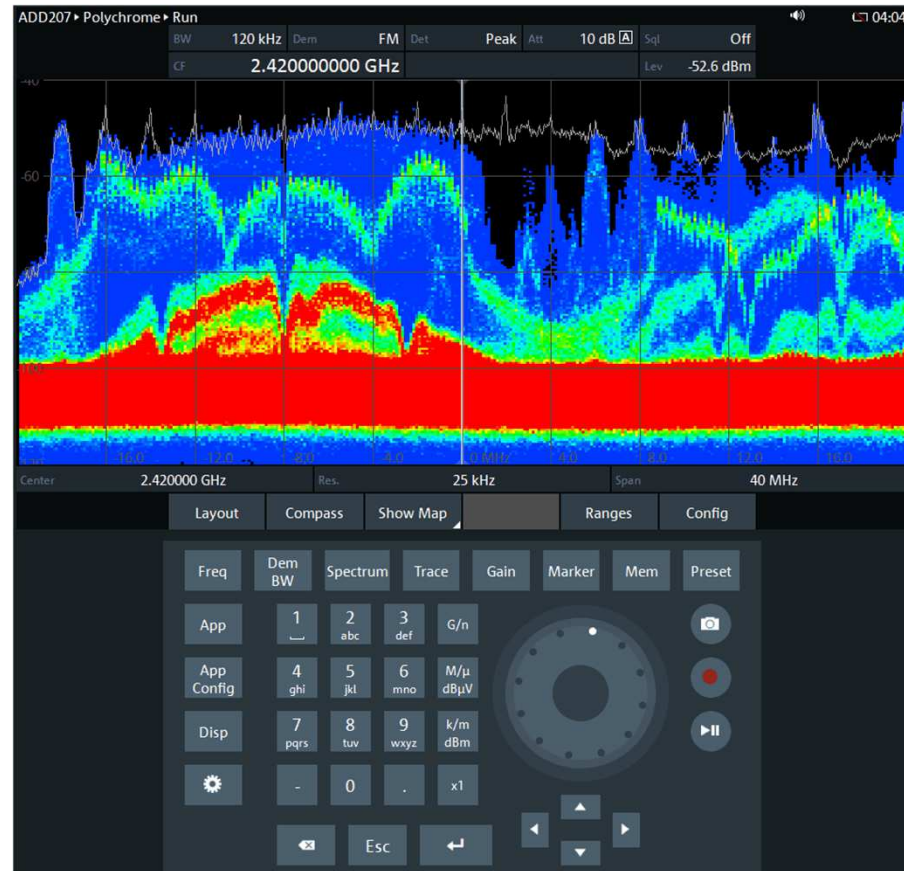
PR200 APPS – POLYCHROME

PR200 APPS

APP

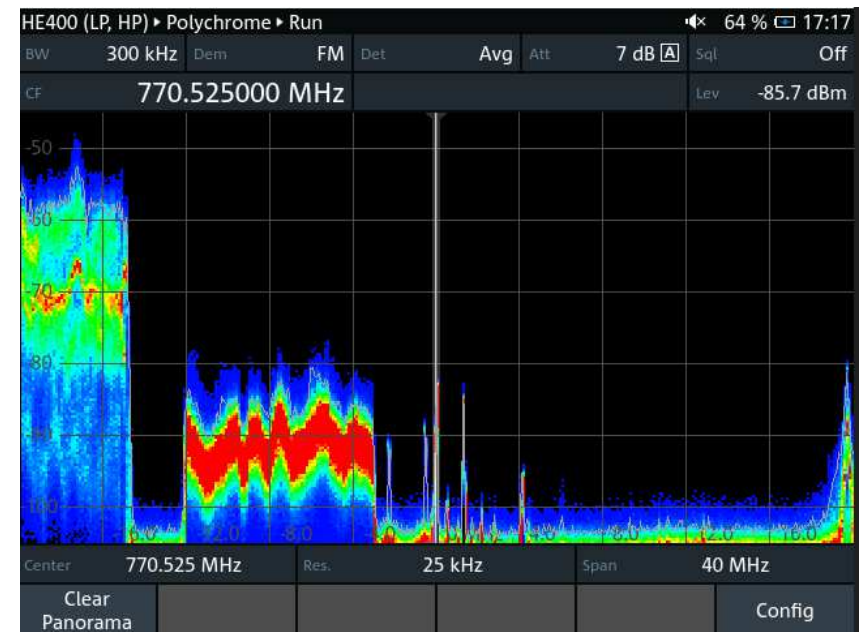
–Polychrome

- ▶ A type of histogram.
- ▶ Uses color to map the relative occupancy of each frequency over time.
- ▶ Show how often the signals appear during the measurement.
- ▶ Color state
 - Red = happens very frequently
 - Orange = happens frequently
 - Yellow = happens (orange > yellow > green)
 - Green = happens infrequently
 - Blue = happens very infrequently



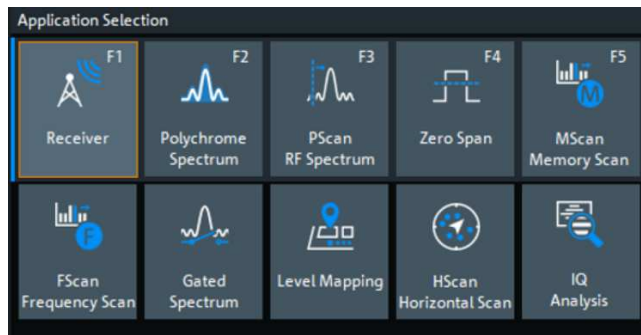
POLYCHROME SETTINGS

- ▶ Polychrome is done in Receiver Mode (40 MHz BW or less)
 - All settings are carried over from Receiver Mode
- ▶ Polychrome specific settings can be accessed using the App Config key
 - Clear Panorama: Clears current display
 - Config:
 - Persistence: How long a colored dot stays before resetting
 - 100% Time: Sets how long a signal level must be active to be given a red dot
- ▶ Polychrome data is not stored in trace recordings
 - Polychrome information is shown in screenshots

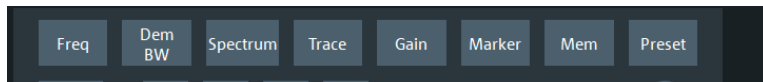


PR200 POLYCHROME

1. Select “Polychrome Spectrum” using the arrow keys or rotary control and press the center of the rotary control or the “Enter” key.

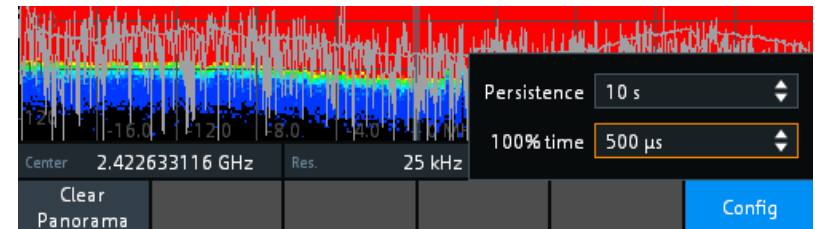


2. Parameters and functions, except the “App Config” portion, can be accessed via the top row buttons as shown in the PR200 Receiver section.

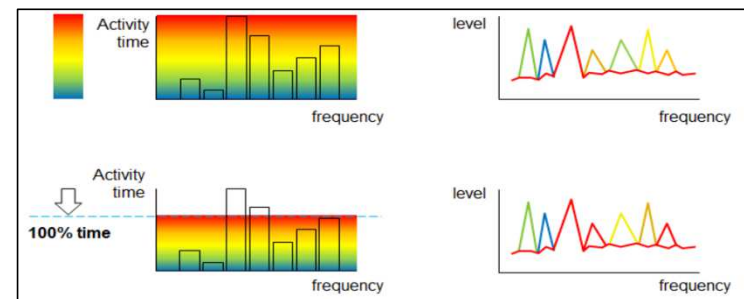


3. Press “App Config” and “Config” to access “Polychrome” specific settings.

1. Select “Persistence” using the rotary control and press the center of the rotary control to enter the “Persistence” value ranging between 10 s to 100 ms. Use a high value for fast signal.

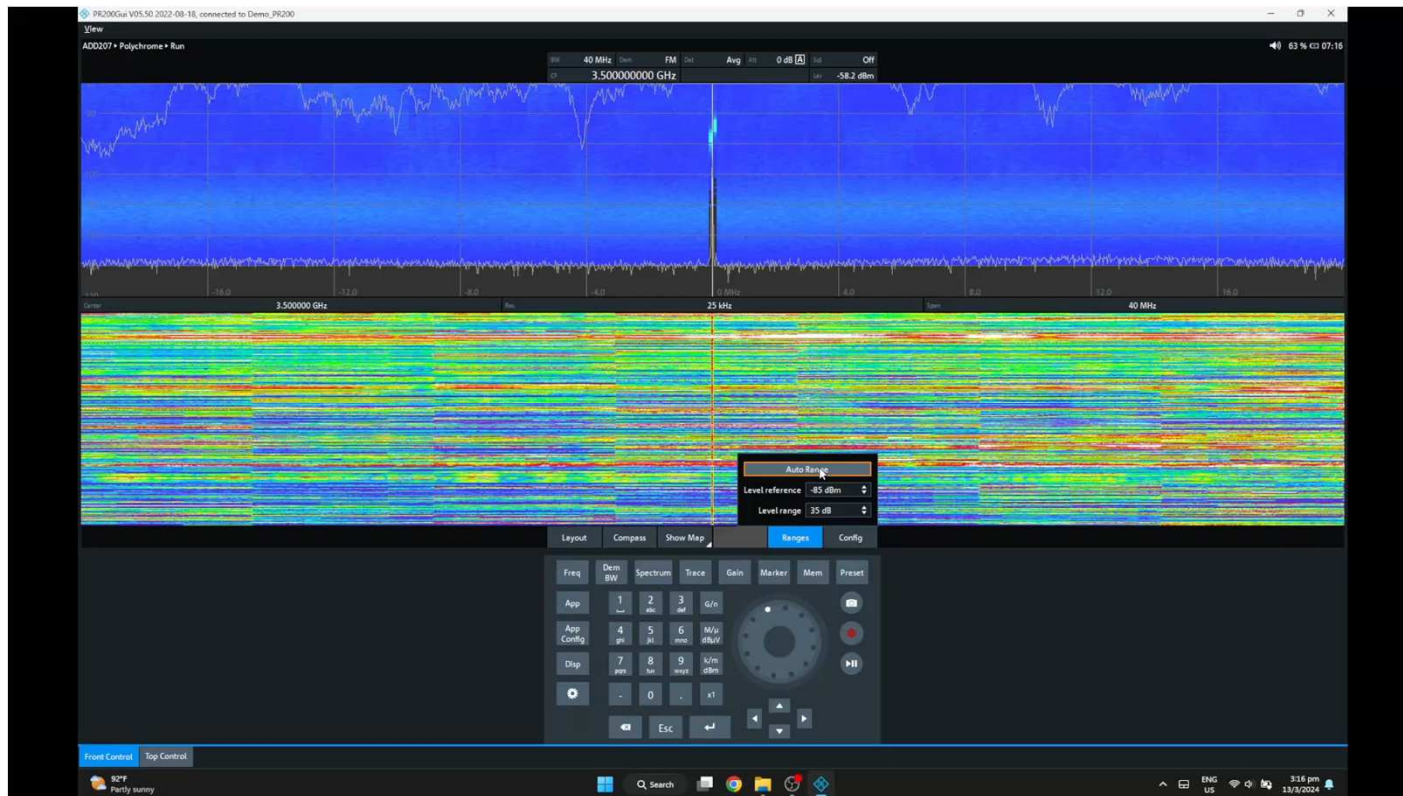


2. Select “100%time’ to enter the activity time ranging between 100 ms to 100 us. Low value will shift fast signal activity towards upper color range (towards red).



PR200 POLYCHROME

- The following is a short example on the use of polychrome to hunt for interference within 5G signals.





HANDS ON

- ▶ Polychrome Touch & Feel
- ▶ Observe 2.4GHz ISM
- ▶ Look for a simulated interference
- ▶ *(est. 15 - 30 min)*



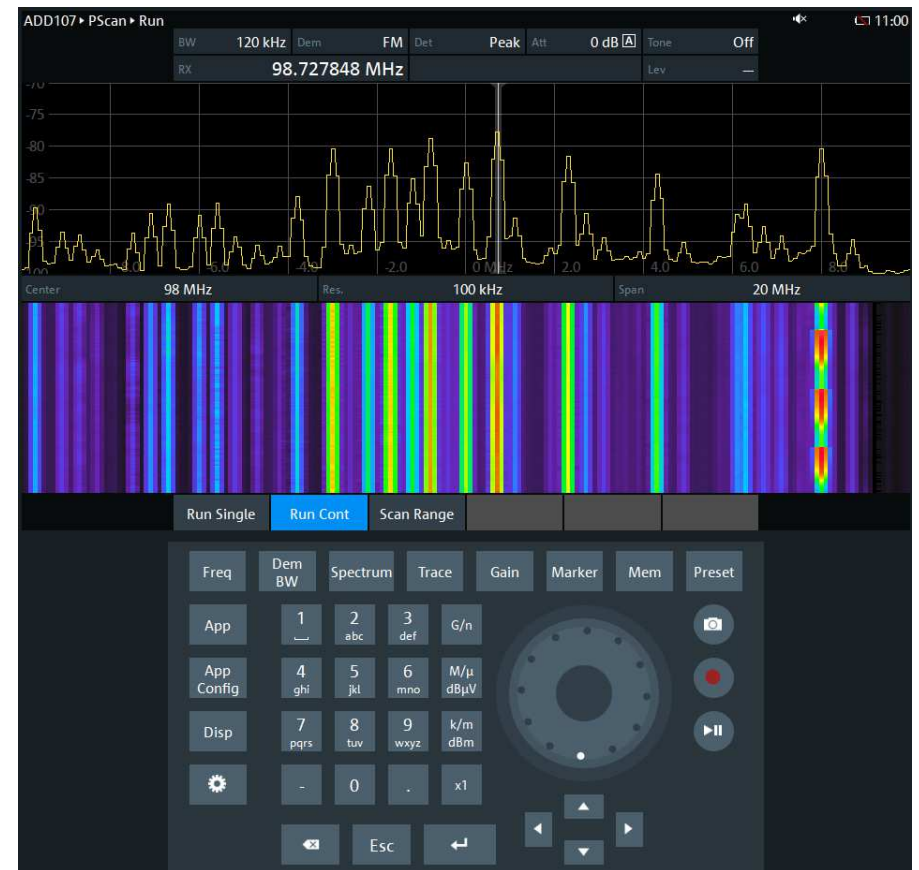
PR200 APPS – PSCAN

PR200 APPS

APP

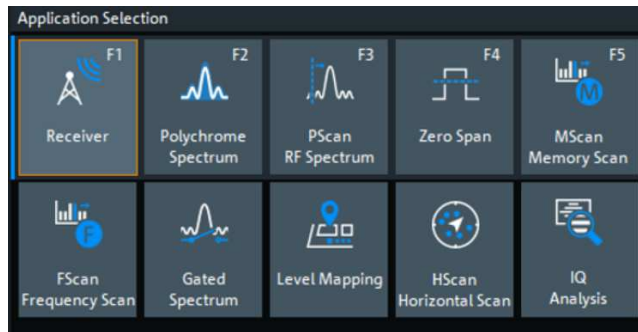
–PScan – “App Config” – “Run Cont.”

- ▶ Perform scan continuously.
- ▶ User cannot listen to the audio while scanning is in progress



PR200 PSCAN

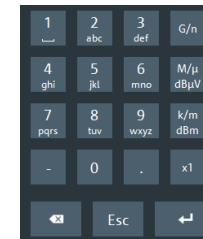
1. Select “PScan RF Spectrum” using the arrow keys or rotary control and press the center of the rotary control or the “Enter” key.



2. Press “Freq” on the top row and follow by the “Start Freq”.



3. Enter the desired start frequency via the number pad and press the “Enter” key or the appropriate units key to accept the entry.

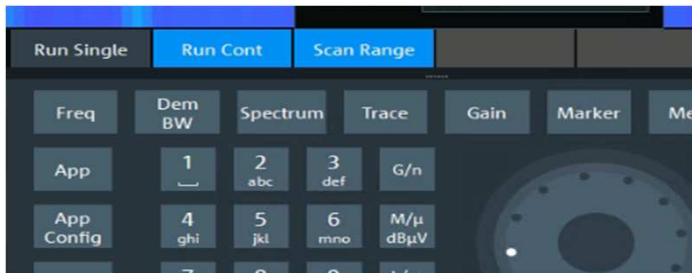


4. Repeat steps 2 to 3 for the “Stop Freq”.
5. Start, stop, center frequencies, resolution and span can also be changed via “App Config” -> “Scan Range” menu.



PR200 PSCAN

6. Audio will be heard only when the scan is stopped.
To stop the scan, press “App Config” and de-activate the scan by pressing “Run Cont”



7. To tune to an active frequency, press “Freq” followed by the “RX Freq”.



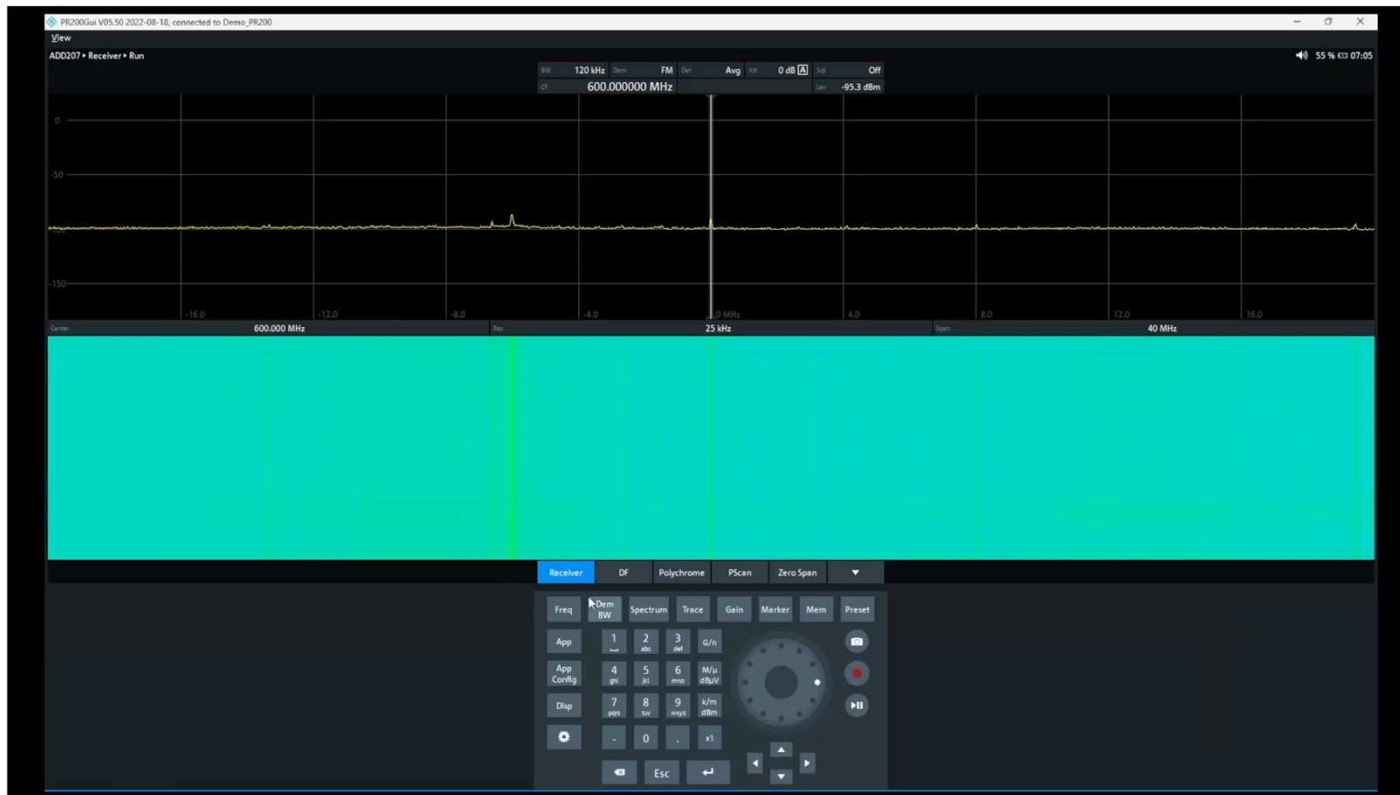
9. Or use the rotary control to scroll through the spectrum.



10. Markers can be enabled by pressing the “Marker” button on the top row as shown in the PR200 Receiver section.
11. Trace can also be recorded using the same method as shown in the PR200 Receiver section.

PR200 PSCAN

- The following is a short example on the use of PScan to hunt for interference within 5G signals.





HANDS ON

- ▶ PScan Touch & Feel
- ▶ Observe 5G Signals live
- ▶ Play audio
- ▶ *(est. 15 - 30 min)*



PR200 APPS – LEVEL MAP

PR200 APPS

APP

–Level Mapping

Displays the levels as a colored track on the map. The track consists of a sequence of colored points. Each point corresponds to

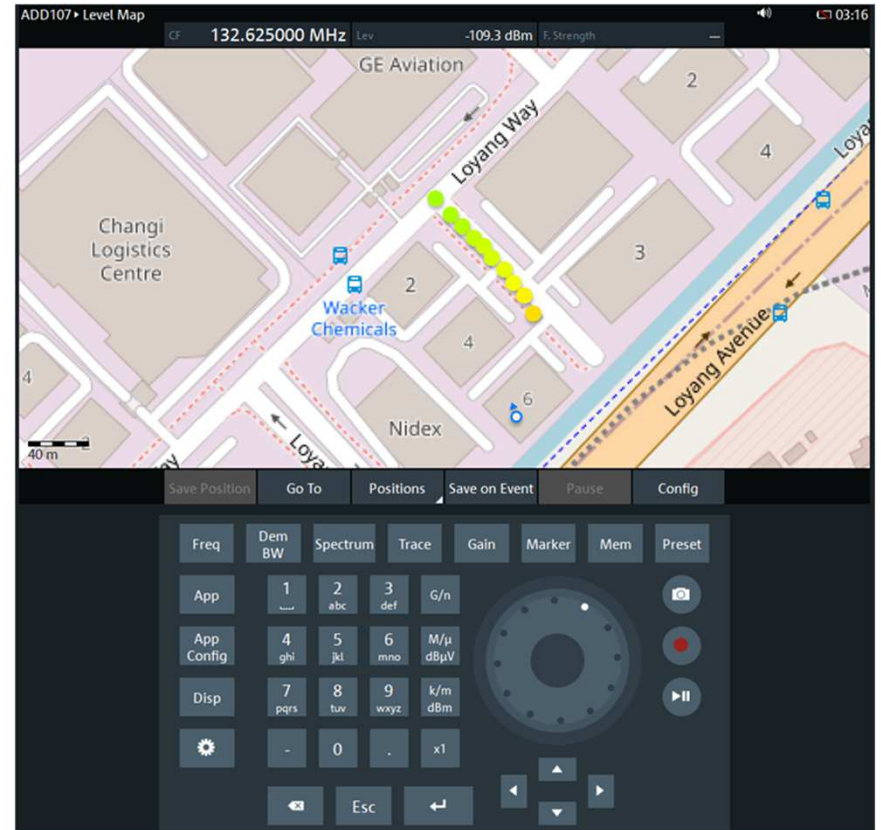
- the location at which the signal was measured. The color depends on the strength of
- the signal at that particular location.

Typical use cases are:

- Coverage measurements
- Transmitter range testing
- Interference hunting

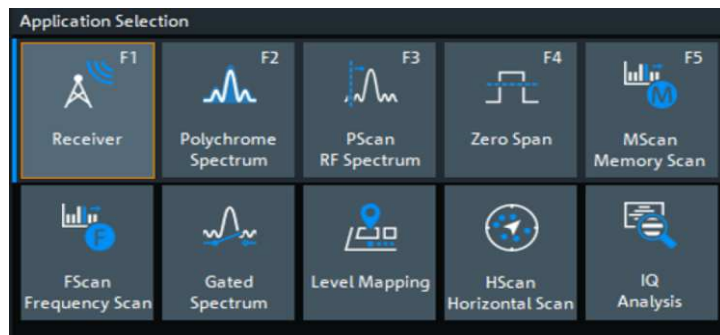
Typical measurement methods are:

- Walking around the area of interest
- Driving around the area of interest

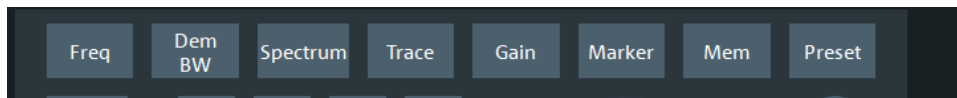


PR200 LEVEL MAPPING

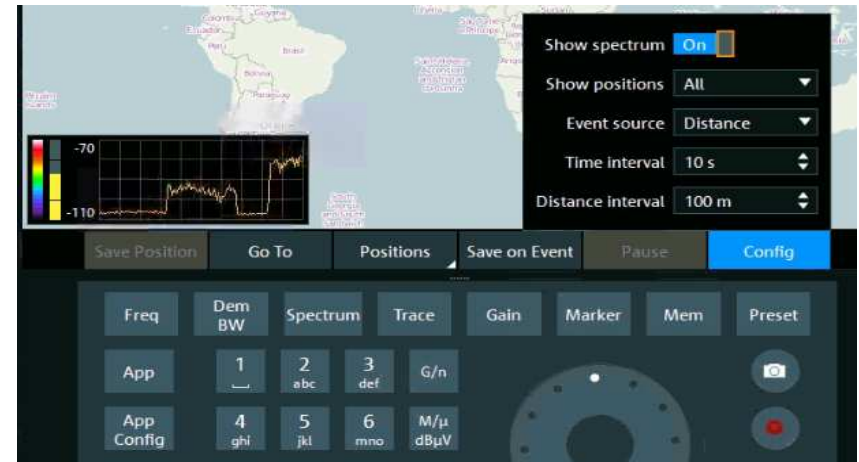
1. Select “Level Mapping” using the arrow keys or rotary control and press the center of the rotary control or the “Enter” key.



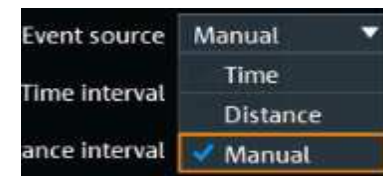
2. Set the desired measurement parameters for the target frequency as shown in the PR200 Receiver section.



3. Press “App Config” follow by “Config”.
4. Toggle “Show spectrum” to open the spectrum view in the map.

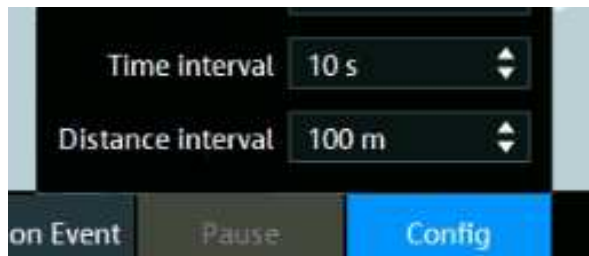


5. Select “Event source” to set the type of event for capturing the position and level.

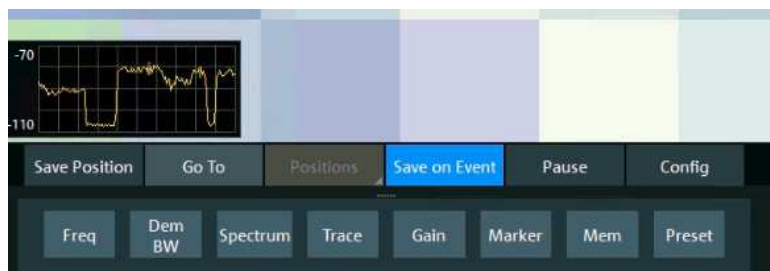


PR200 LEVEL MAPPING

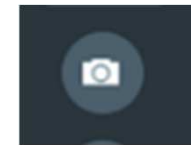
- Depending on the type of event source, the corresponding “Time interval” or “Distance interval” will be considered.



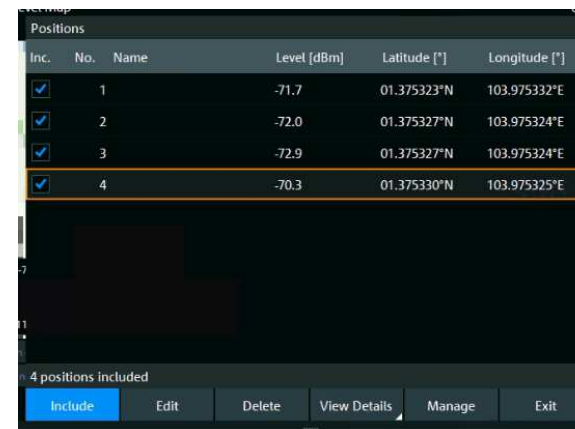
- If “Manual” is chosen as the “Event source”, the “Save Position” button will be active for manual saving of position and level.



- To take a snapshot of the current display, press the button with camera icon. Image will be saved in “..\PR200\Screenshots” folder.



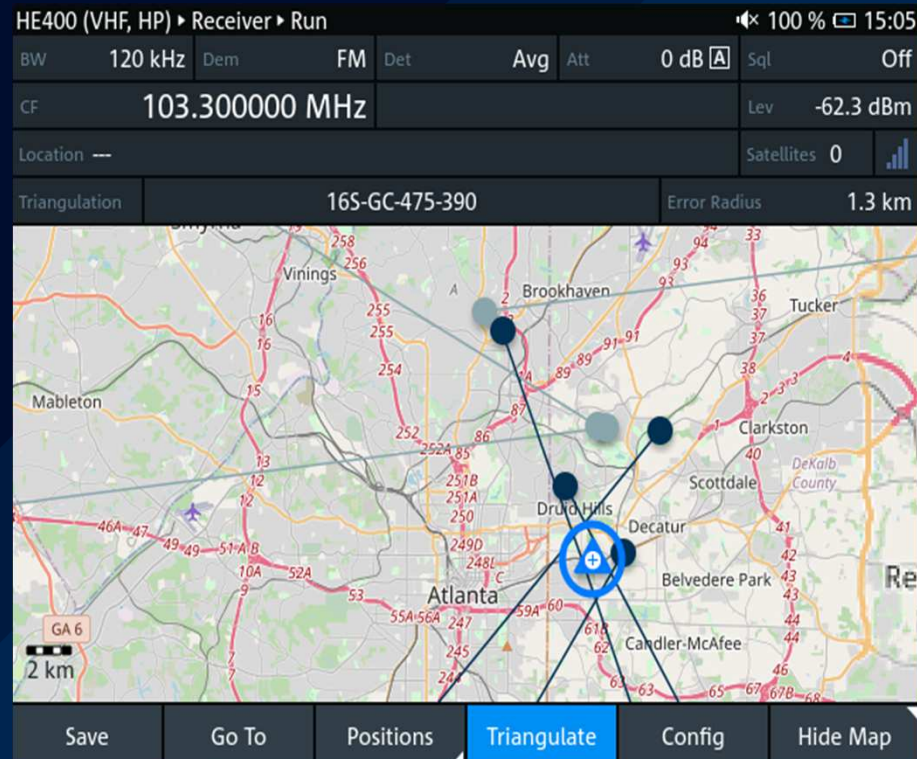
- To view the list of positions saved, press “Positions” follow by “Show List”. The list can be managed in the dialog that opens.

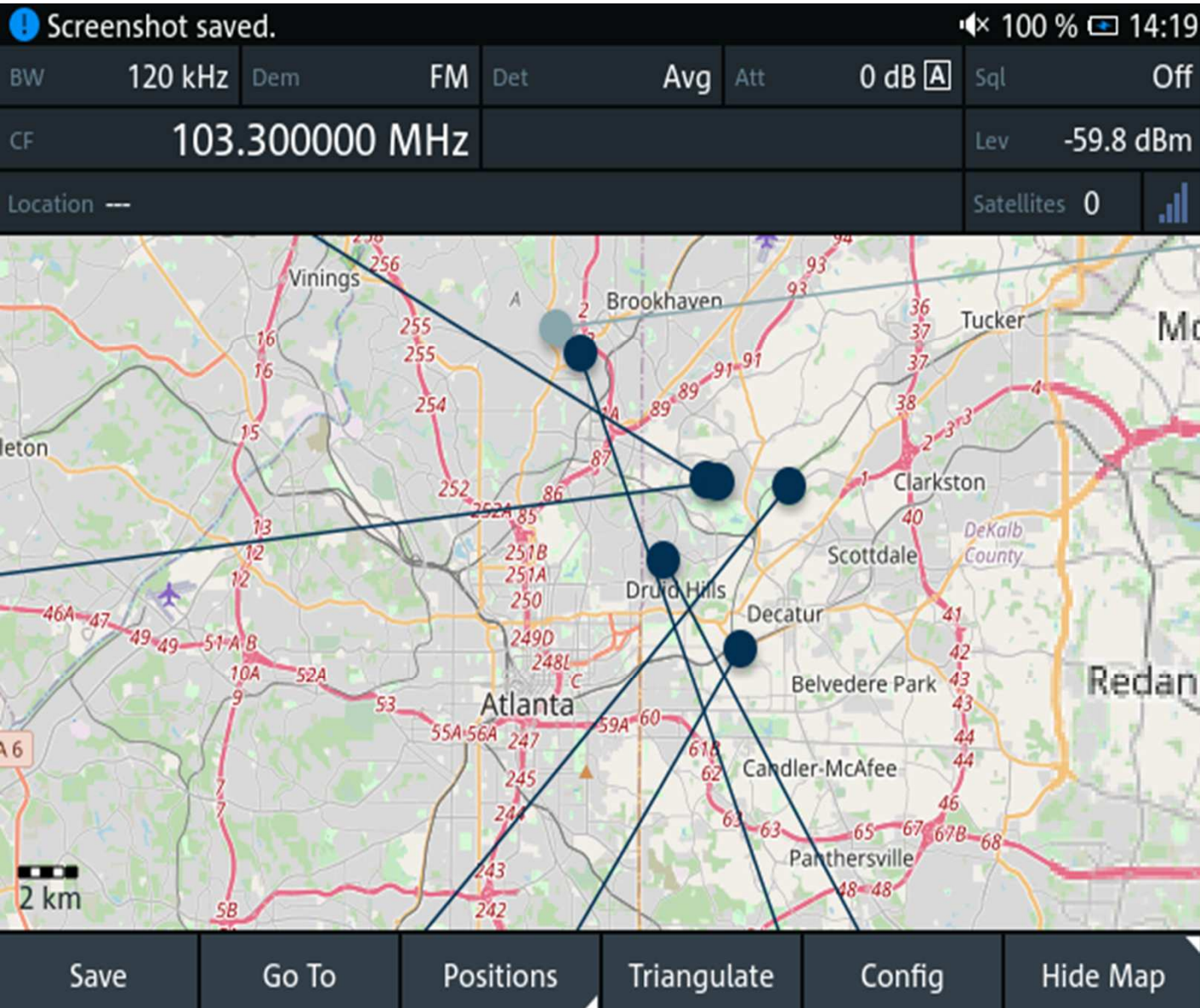


TRIANGULATION

ROHDE & SCHWARZ

Make ideas real





TRIANGULATION PROCESS (CONT.)

- ▶ 3) Save the azimuth:
 - Disp -> Show Map -> Save -> Current Position
 - Single Press of HE400 Trigger
 - Double Press of HE400 Trigger (HSCAN)
- ▶ 4) Repeat steps 2 and 3 at all previously identified locations

100 % 15:04

Sql	Off
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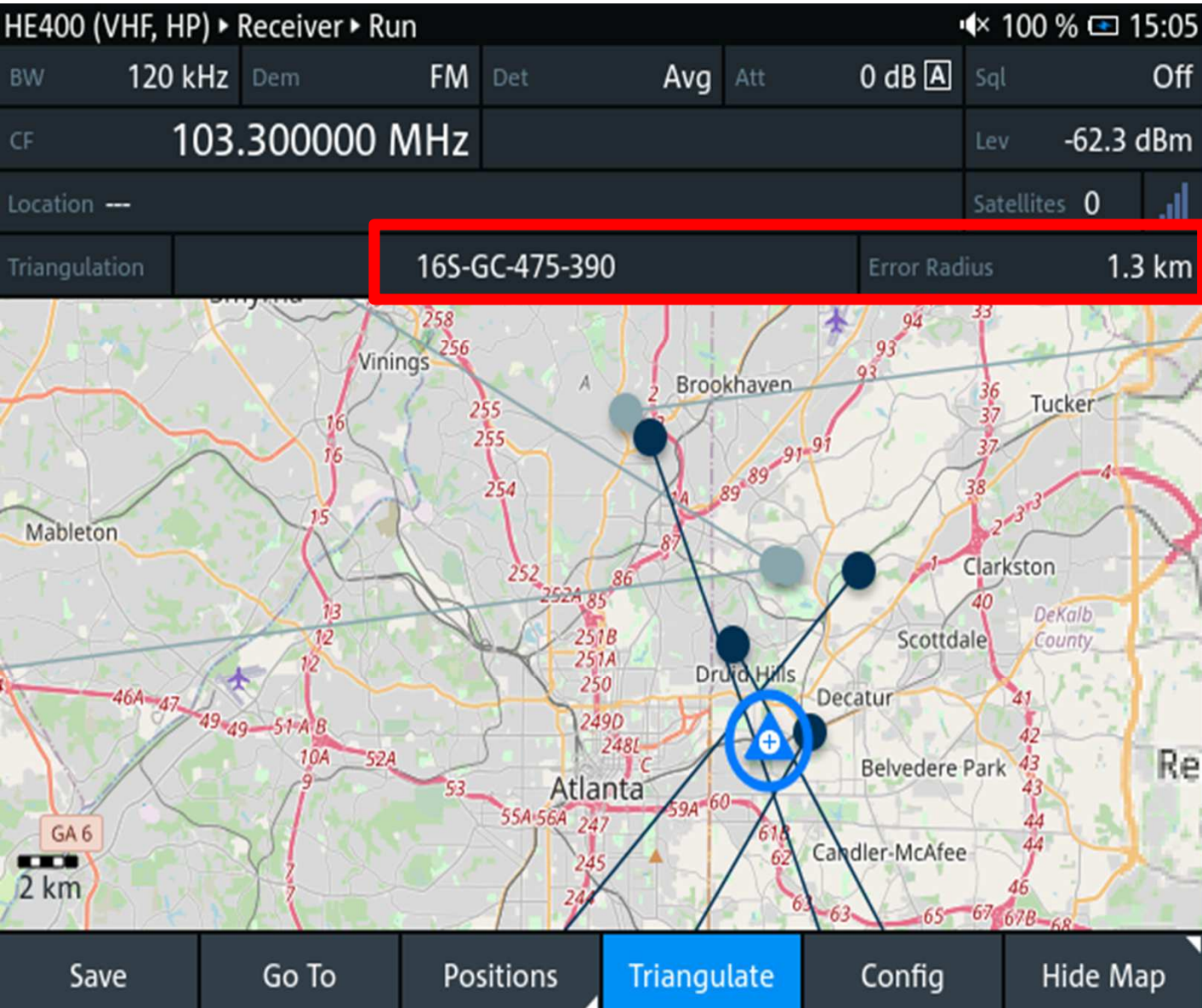
Lev	-61.2 dBm
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Satellites 0 

Hide Map

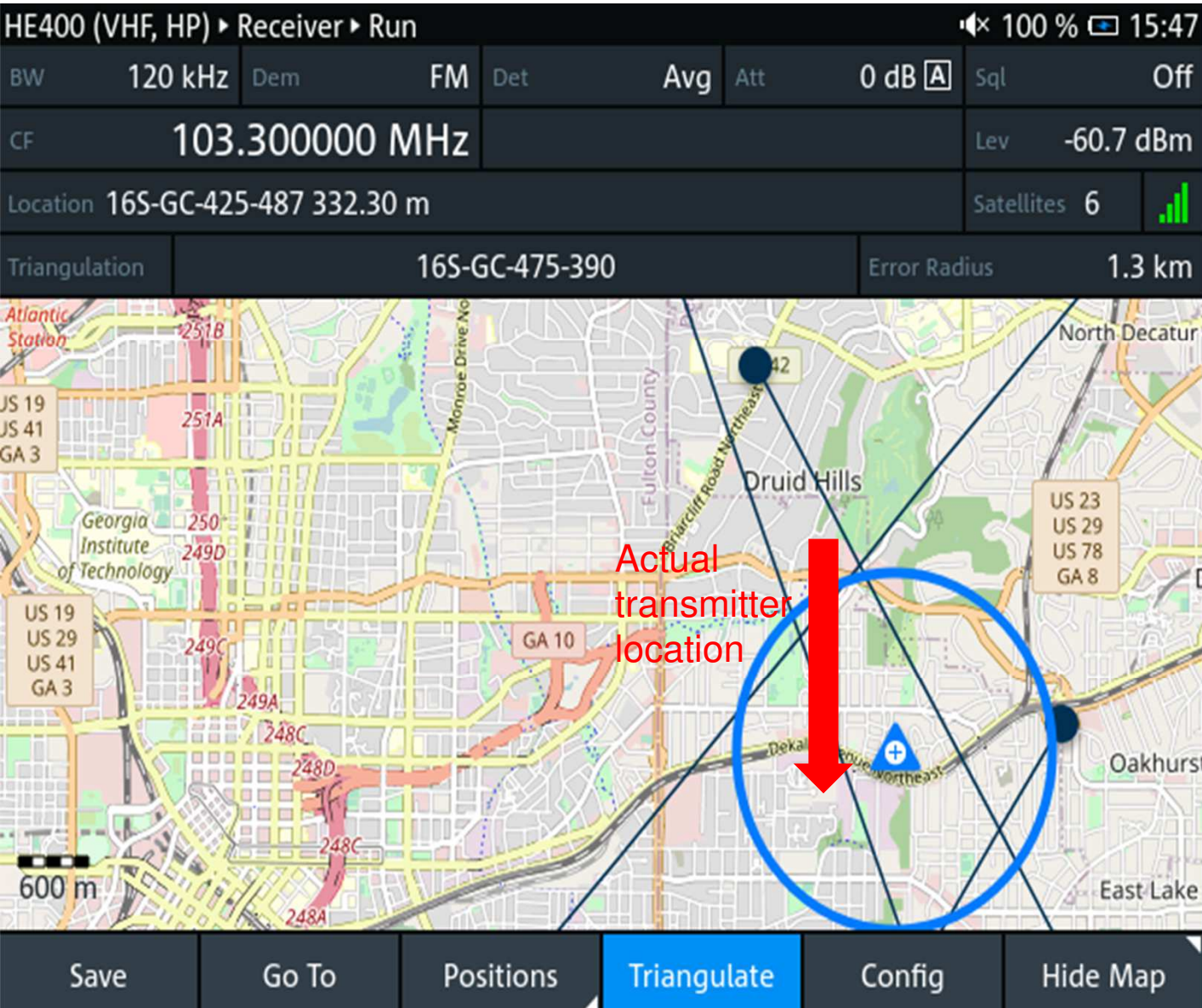
TRIANGULATION PROCESS (CONT.)

- ▶ 5) Throw out “outlier” bearings. These bearings may be erroneous or correspond with real reflections. This can be a judgement call. RF is messy!
 - When viewing the map press Positions (F3) then select the outliers using the arrow keys. Include or exclude them in your triangulation by using the F1 key



TRIANGULATION PROCESS (CONT.)

- ▶ 6) Triangulate (F4)
 - Geolocation result and error radius are given
 - Requires at least 3 bearings



TRIANGULATION PROCESS (CONT.)

- ▶ In this example the triangulation was off by ~2,000 ft.
- ▶ More LOBs would likely improve the accuracy



PR200 APPS – MSCAN

PR200 APPS

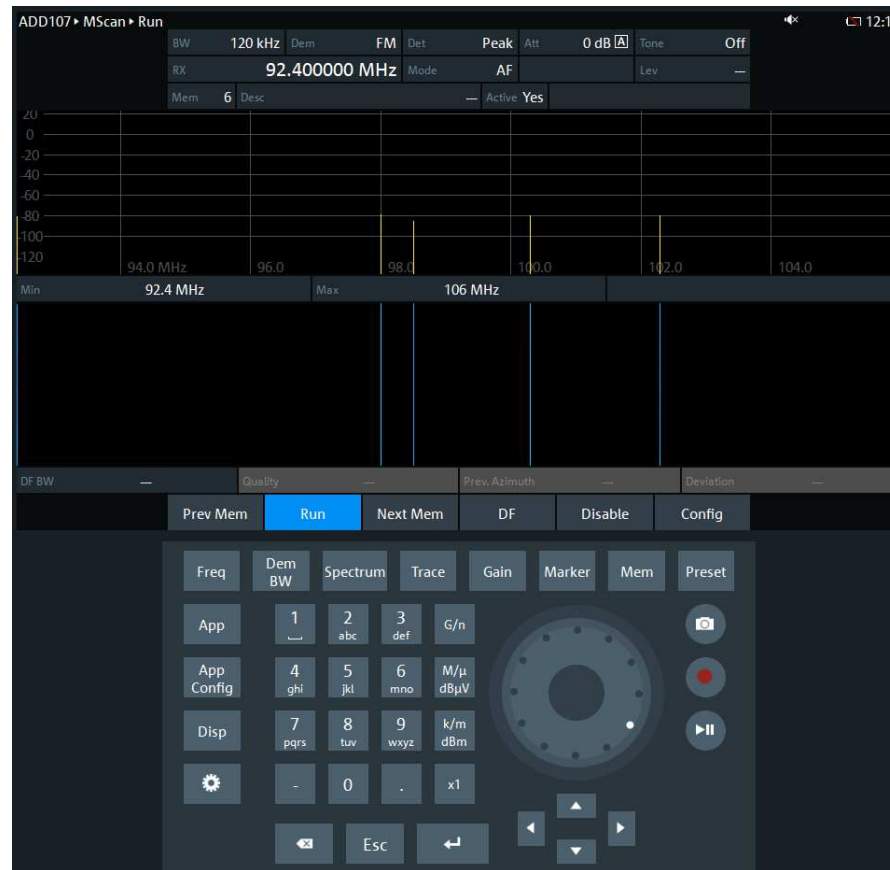
APP

–MScan

- ▶ Perform scan of channels in the memory list.
- ▶ Channels must be entered into memory list by using “Mem” – “Add Entry” or using “Import” in the “Mem” – “Mem List” – “Manage” key.
- ▶ Memory list can be prepared by using InstrumentView software.

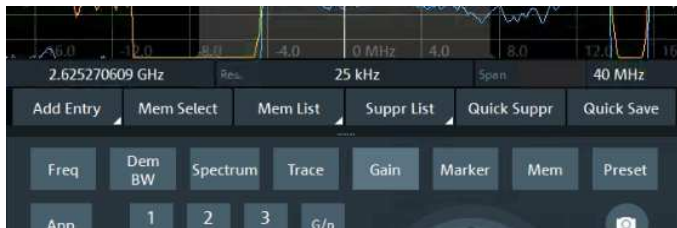
Typical uses of memory list are:

- ▶ **Automatic scanning** of the stored signals (with audio demodulation) via the "MScan" app.
- ▶ **Manual selection** of specific signals from the list via the [Mem] key.
- ▶ Up to 9999 entries can be saved in this list. However, only 999 entries are saved persistently.

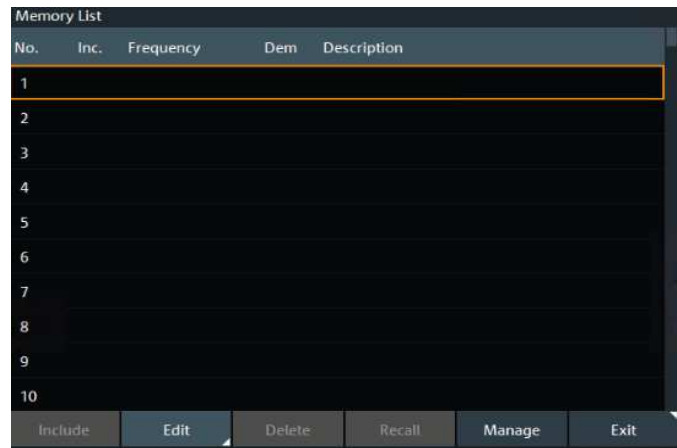


PR200 MScan

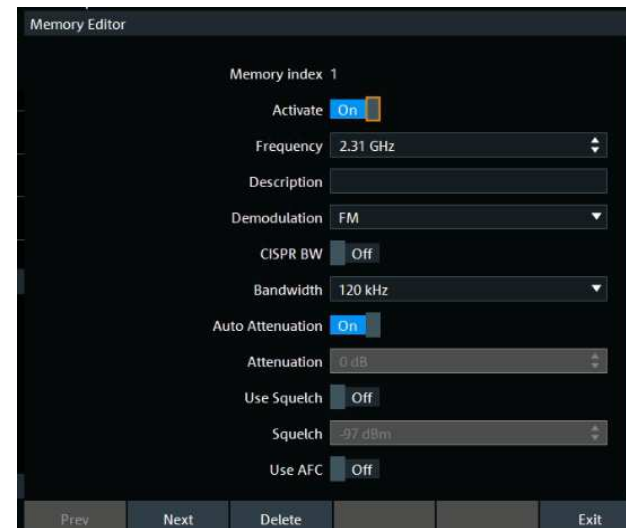
1. Add entry into memory list. Press “Mem” button.



2. Press “Mem List” and select an empty row.

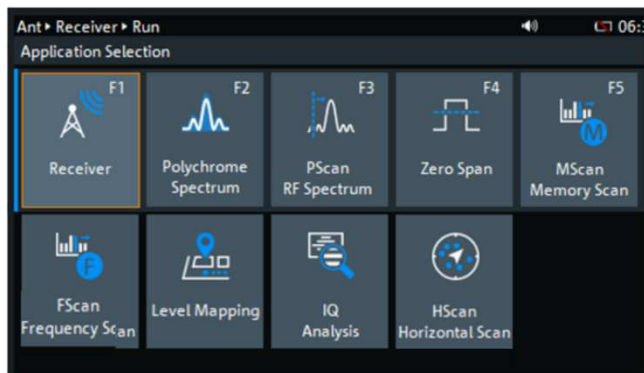


3. Press “Edit” and enter the parameters..Toggle “Activate” to “On” to include it for MScan. Press “Next” to enter the next frequency parameters.



PR200 MSCAN

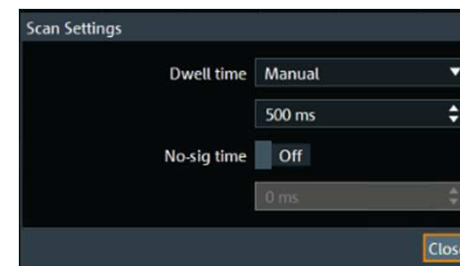
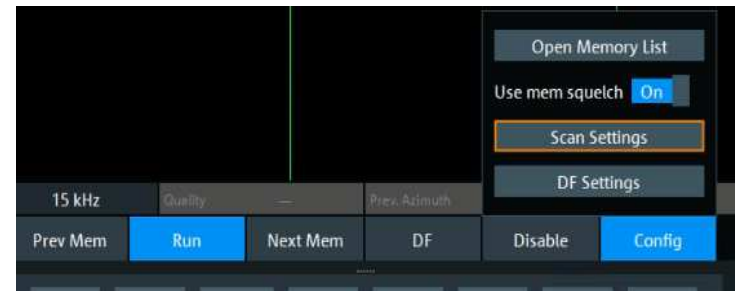
4. Select “MScan Memory Scan” using the arrow keys or rotary control and press the center of the rotary control or the “Enter” key.



5. Press “App Config” and then “Run” to start the scan.



6. Press “App Config” follow by “Config” to change MScan specific settings.





HANDS ON

- ▶ MScan Touch & Feel
- ▶ Enter a list of frequencies
- ▶ Conduct a scan of the list
- ▶ *(est. 15 - 30 min)*



PR200 APPS – HSCAN

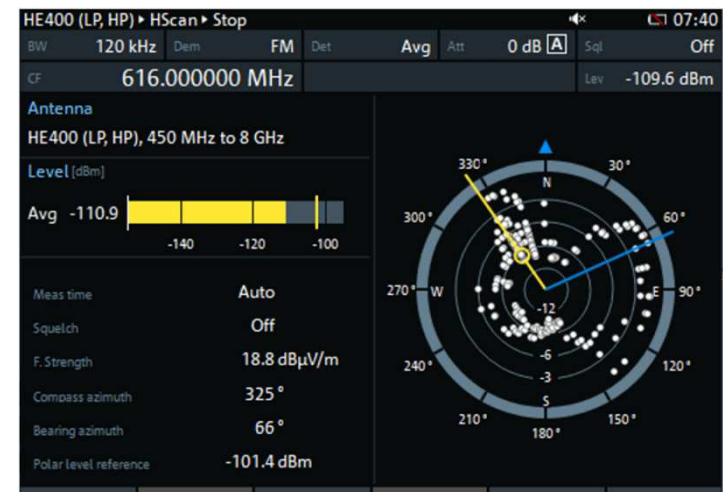
PR200 APPS

APP

–HScan

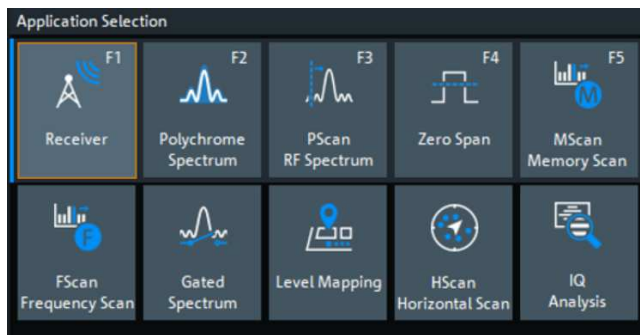
Provide a representation of the power level detected in the receiver over different azimuths which helps to estimate the bearing of the signal.

This is useful when there is more than one main angle of incidence of the signal, which can be caused by reflection or interference. The secondary lobes in the reception pattern (caused by reflection and interference) can be easily identified.



PR200 HSCAN

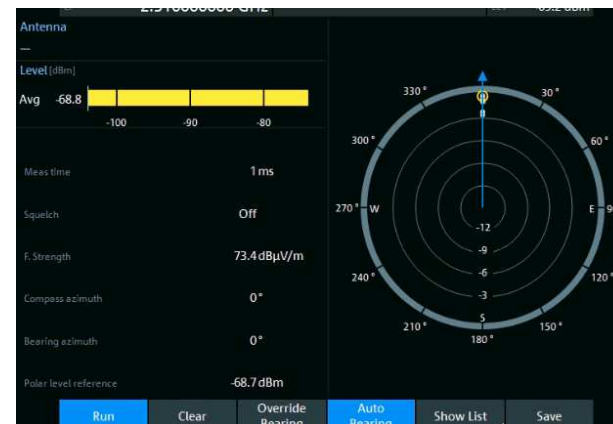
1. Select “HScan” using the arrow keys or rotary control and press the center of the rotary control or the “Enter” key.



2. Set the measurement time to 200 ms (recommended) via “Dem BW” -> “Config” -> “Meas. time”.



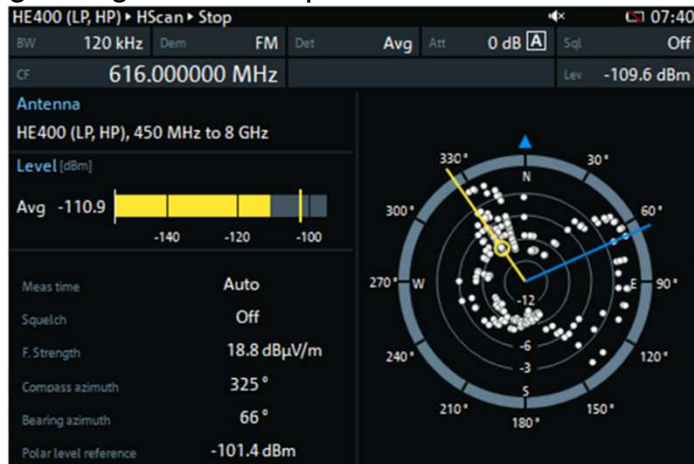
3. Press the “App Config” button, “Auto Bearing” and then “Run”.



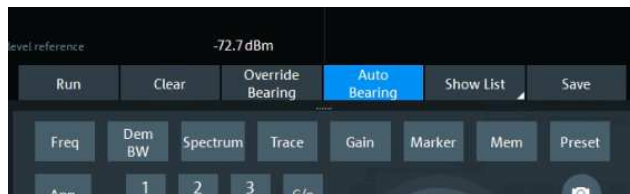
4. Rotate the directional antenna 360 degree at a stationary point until a blue line appears in the polar diagram. Ensure that the HE400 antenna is held at the appropriate polarization.

PR200 HSCAN

- Blue line is autobearing in the direction of the strongest signal level captured.



- To clear the plot, press “App Config” and “Clear”. Press “Run” to repeat the measurement.



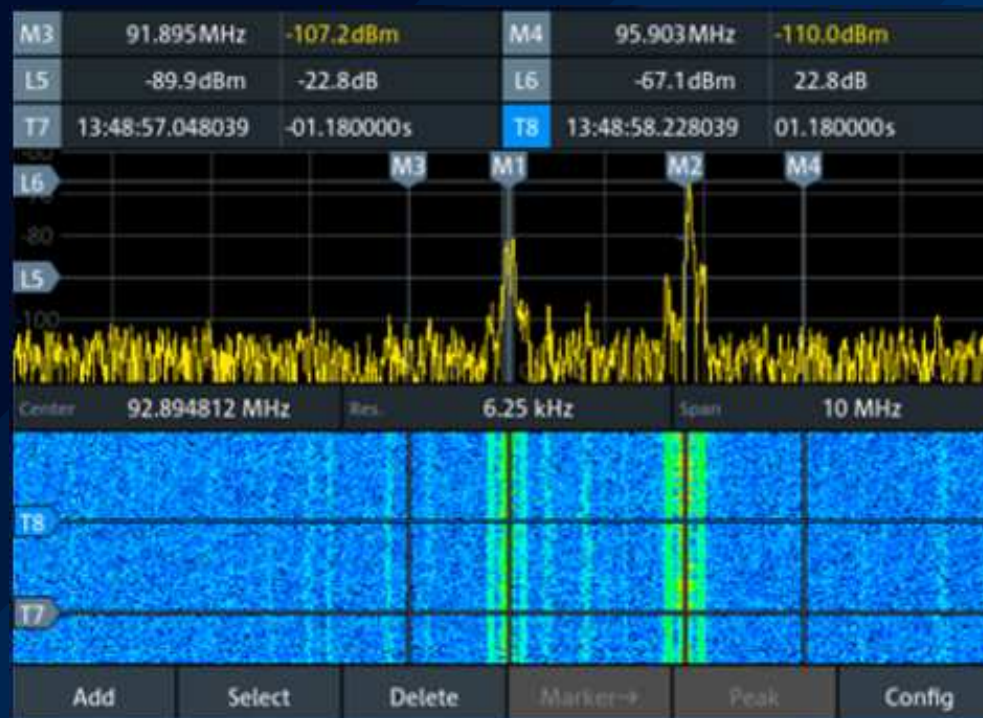
- If required, the current compass direction can override the auto bearing result by pressing “App Config” follow by “Override Bearing”.
- Current bearing and position can be saved by pressing “App Config” follow by “Save”.
- A “.csv” file containing the plot data will also be saved in the “Positions” folder.
- Operation with HE400 antenna trigger button:
 - Single-click: Trigger the run/pause of a scan.
 - Double-click: Save current position, polar diagram, and measurements.
 - Click-and-hold (available only with R&S HE400CEL): Use polar nulls for bearing estimation.



HANDS ON

- ▶ HScan Touch & Feel
- ▶ Interference Hunting Simulation with HScan
- ▶ *(est. 15 - 30 min)*

MARKERS AND LINES



ROHDE & SCHWARZ

Make ideas real





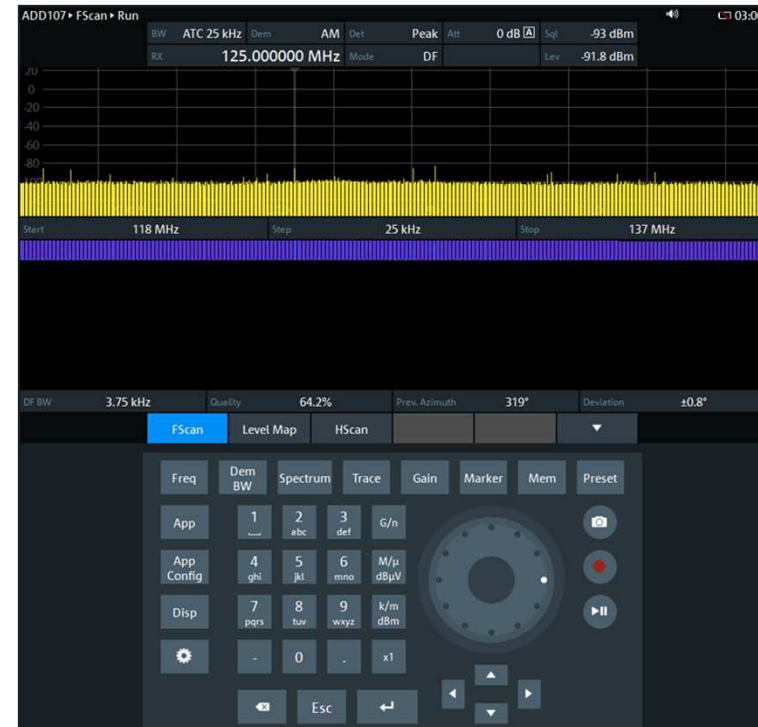
PR200 APPS – FSCAN

PR200 APPS

APP

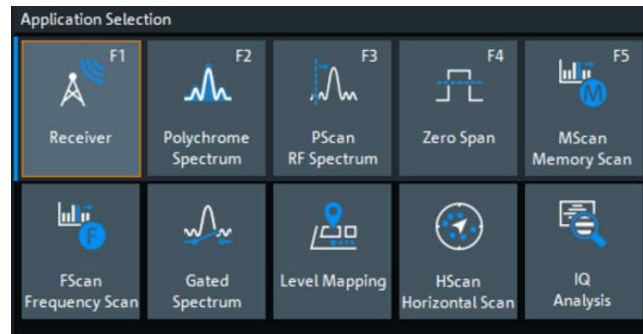
–FScan

"FScan" is typically used for detecting occupied channels within a specific frequency range.

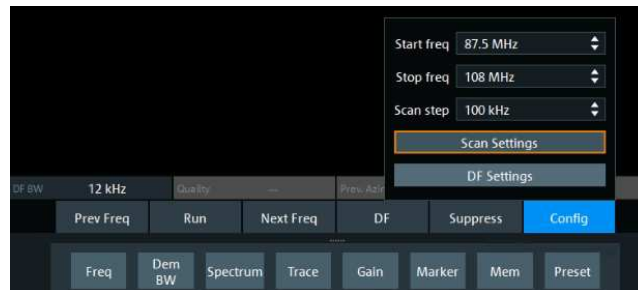


PR200 FSCAN

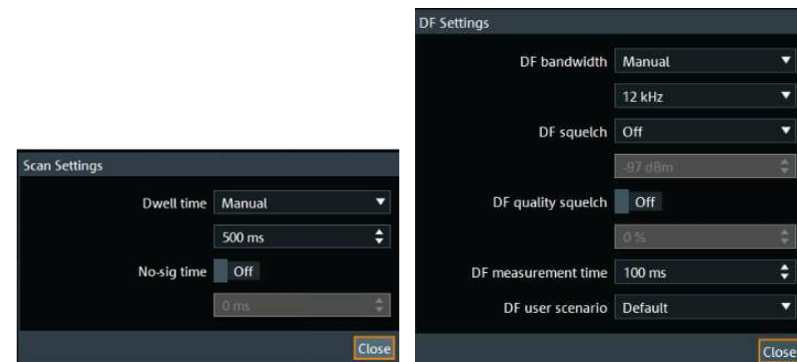
1. Select “FScan Frequency Scan” using the arrow keys or rotary control and press the center of the rotary control or the “Enter” key.



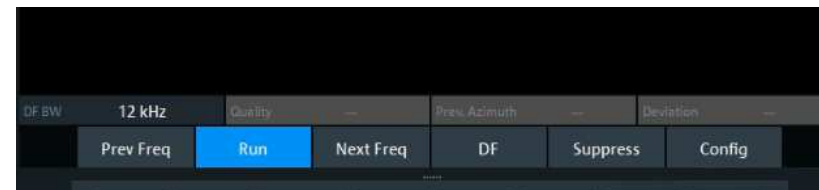
2. Press “App Config” on the top row and follow by the “Config”.



3. Enter the desired start, stop and scan step frequencies.
4. Change Scan or DF settings.

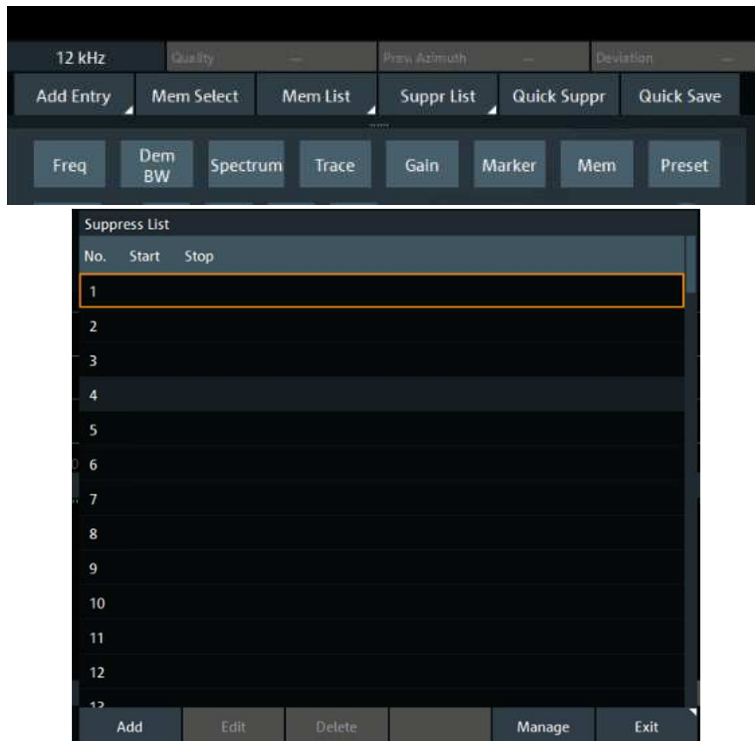


5. Press “Run” to start the scan.

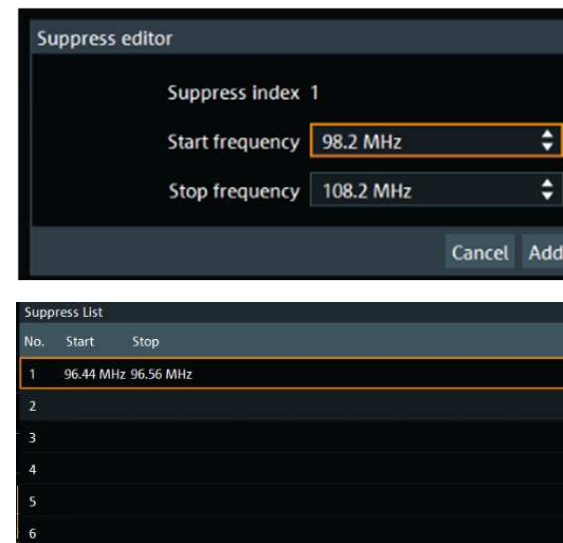


PR200 FSCAN

6. Frequency ranges that are not of interest may be suppressed, to suppress these ranges, press “Mem” on the top row follow by “Suppr. List”.



7. Press “Add” which will bring up the editor. Enter the “Start frequency” and “Stop frequency” range and press “Add”.



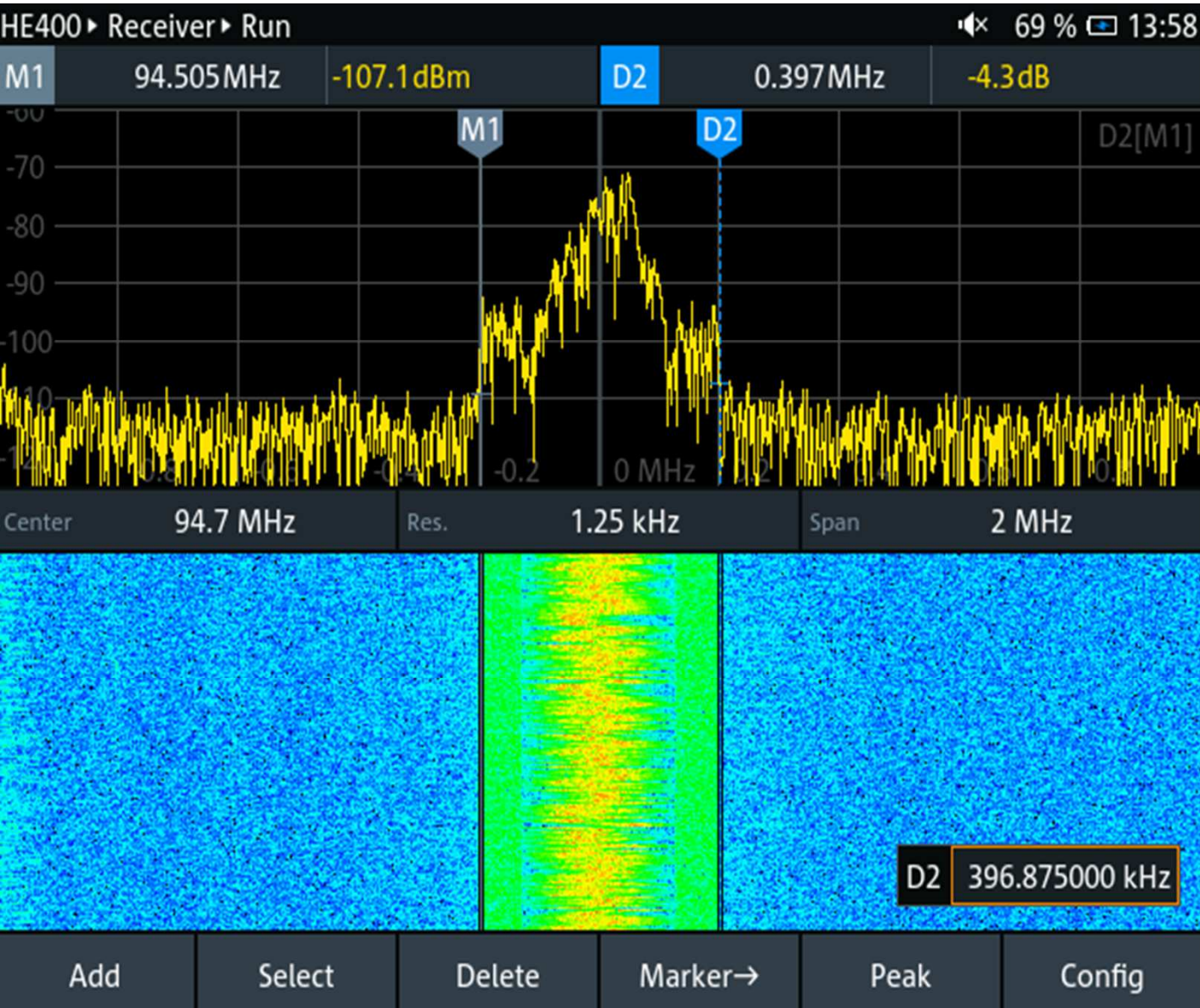


HANDS ON

- ▶ FScan Touch & Feel
- ▶ Observe 5G Signals
- ▶ *(est. 15 - 30 min)*

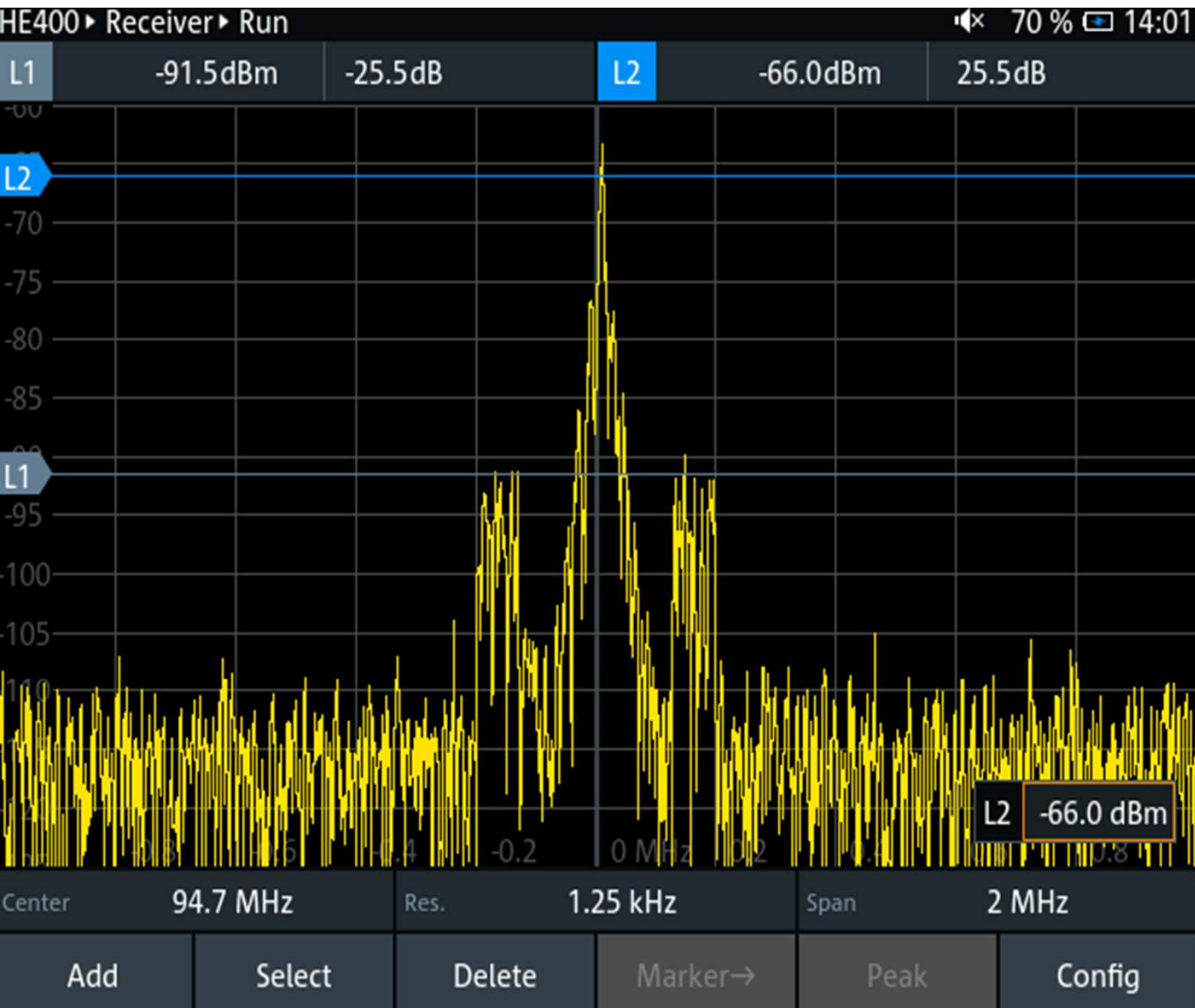
MARKERS AND LINES

- ▶ Used to determine frequency, amplitude, and time values in spectrum and waterfall views
- ▶ Markers and Lines can be accessed in any mode by selecting the “Marker” key
- ▶ Markers:
 - 6 Markers available
 - Allows user to identify the center frequency of a signal
 - Useful for Evaluating / Measuring absolute frequency and frequency deltas (bandwidth)
- ▶ Lines:
 - 2 Lines available
 - Measures amplitude in spectrum display
 - Measures time in waterfall display
- ▶ Delta (difference) between markers and lines is also calculated and displayed.



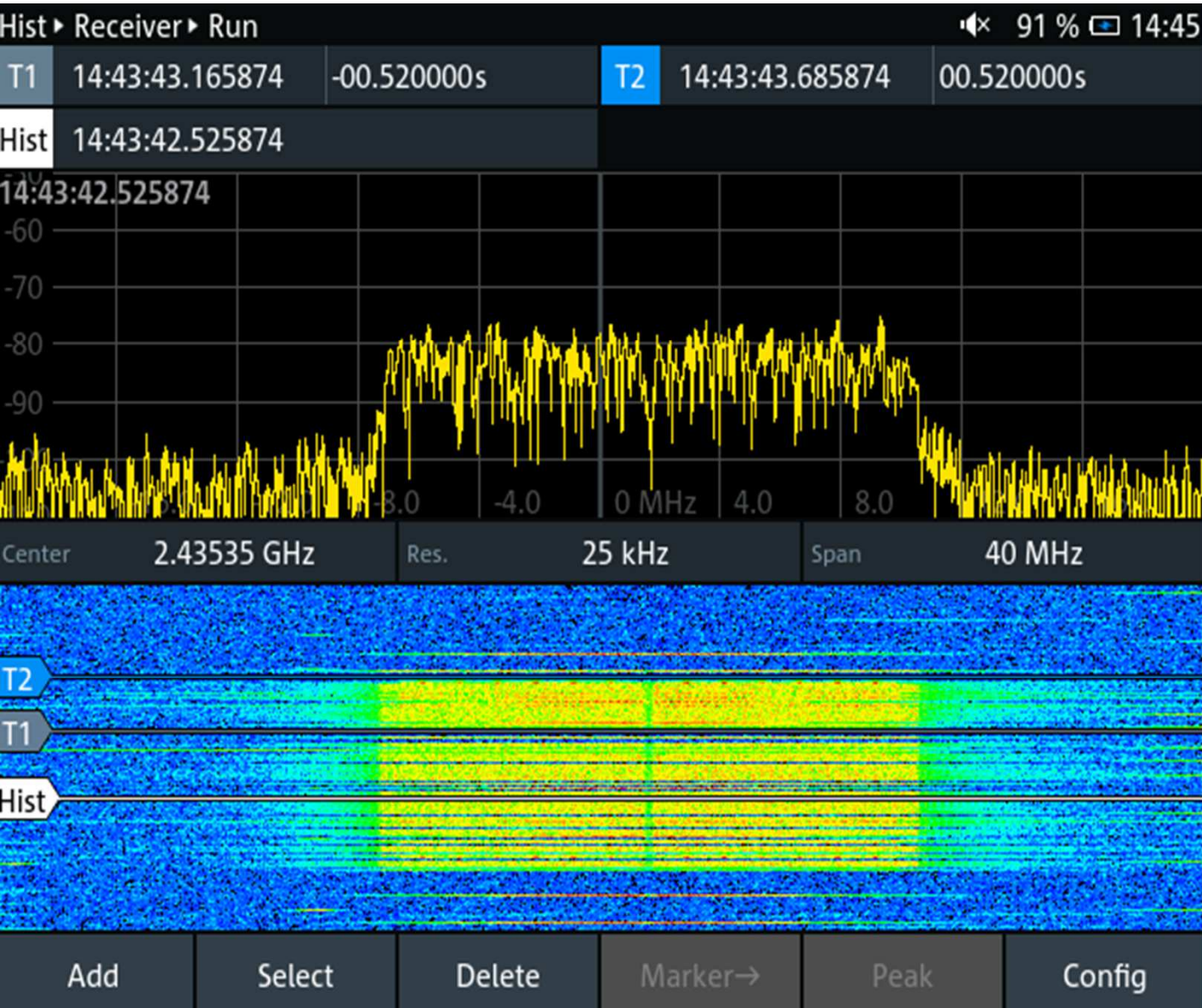
MARKER EXAMPLE

- ▶ M1 Provides Frequency and Amplitude (94.505 MHz @ 107.1 dBm)
- ▶ D2 Measures Frequency and Amplitude difference (.397 MHz and 4.3 dB)
- ▶ 3 Markers with 3 Deltas possible or 6 individual markers
- ▶ Valuable to identify signal bandwidth or frequency spacing



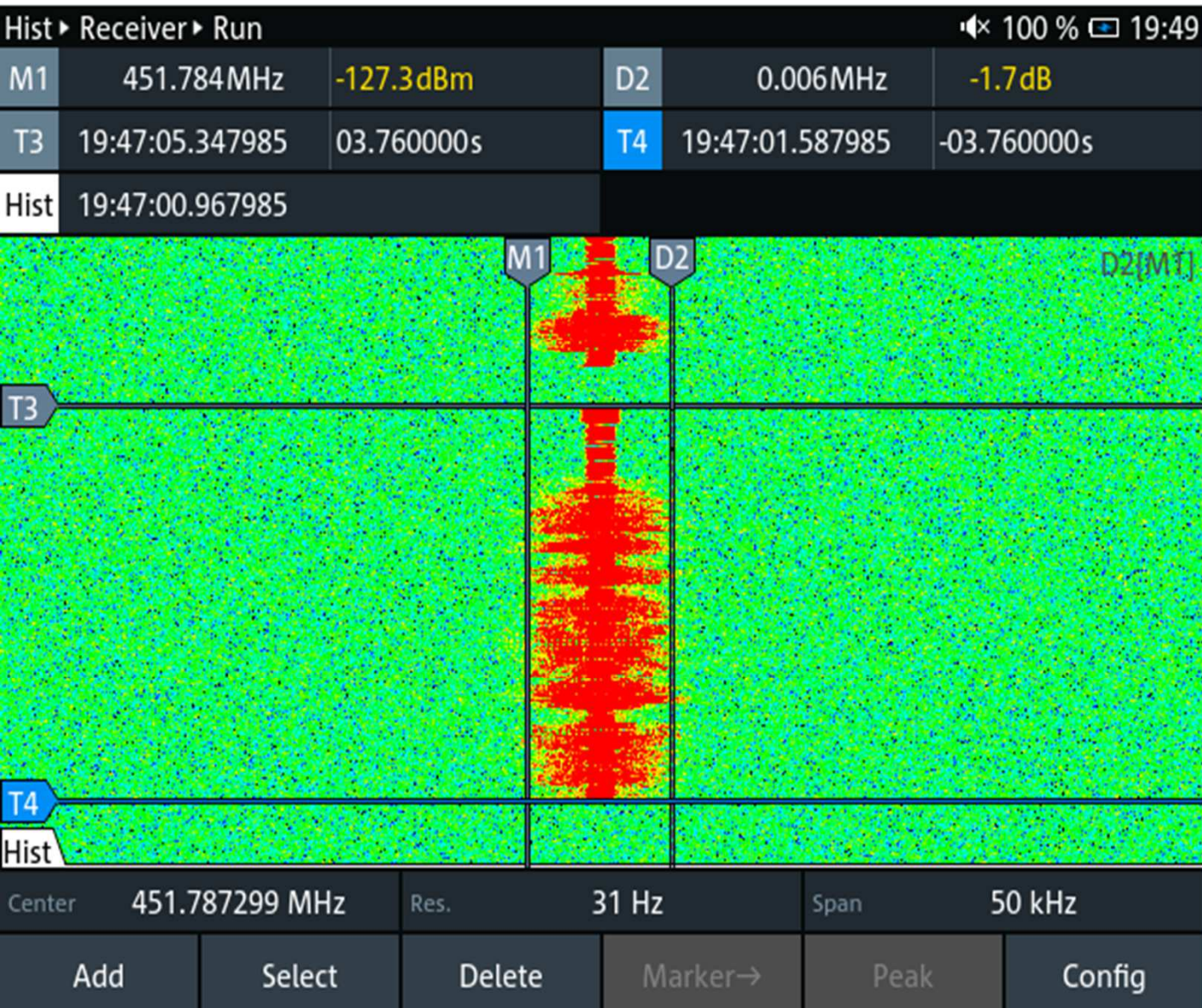
LINES EXAMPLE SPECTRUM

- ▶ L1 Provides signal amplitude and difference (-91.5 dBm / 25.5 dB below L2)
- ▶ L2 also provides signal amplitude and difference (-66 dBm / 25.5 above L1)
- ▶ Valuable to determine signal strength over noise floor or relative to other signals



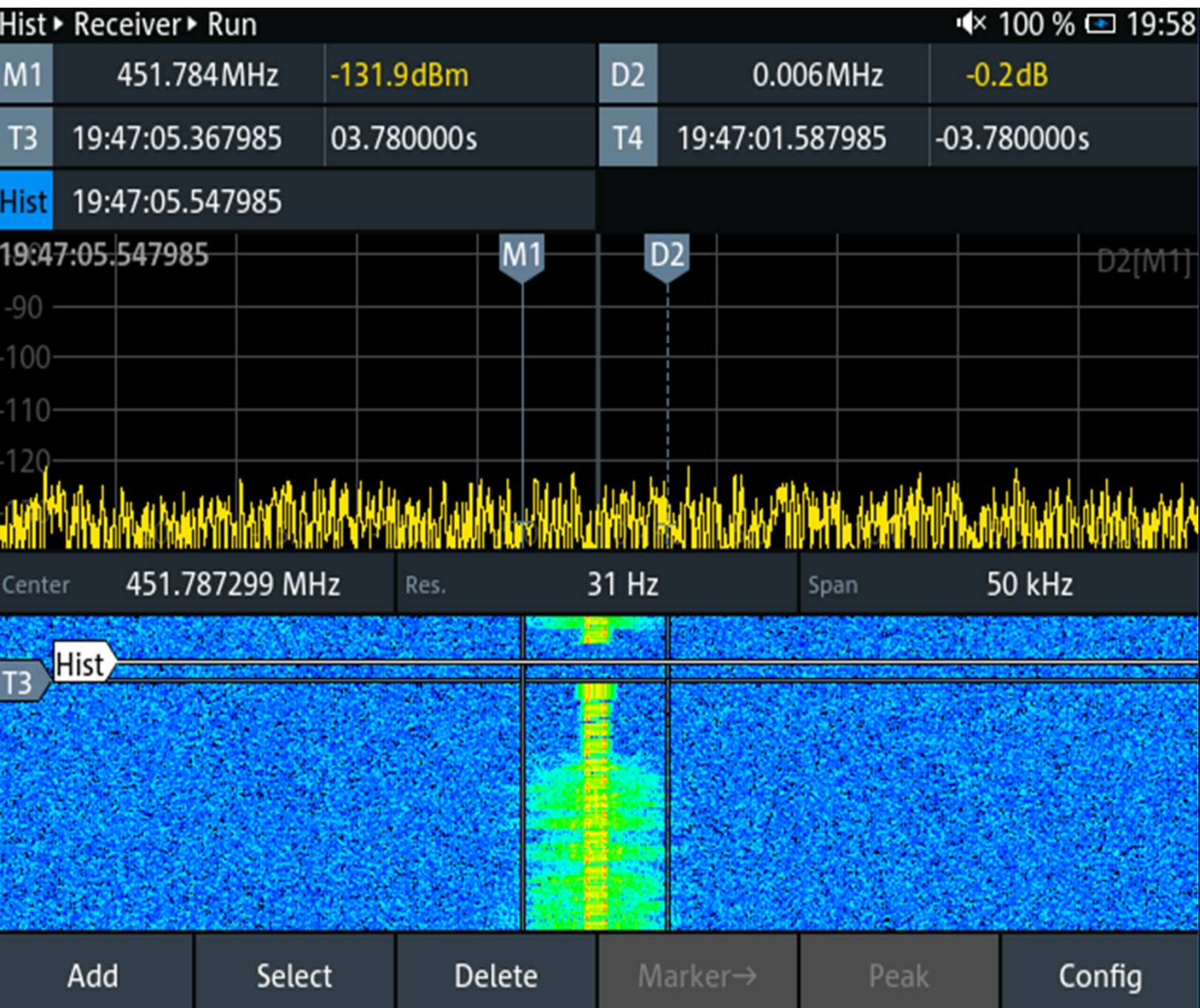
LINES EXAMPLE WATERFALL

- ▶ T1 shows selected time and difference (14:43:43.16 / 0.52 seconds before T2)
- ▶ T2 shows selected time and difference (14:43:43.68 / 0.52 seconds after T1)
- ▶ Valuable to determine signal duration, patterns in duration, etc
- ▶ Markers can be used within waterfall, but only if spectrum display is selected.



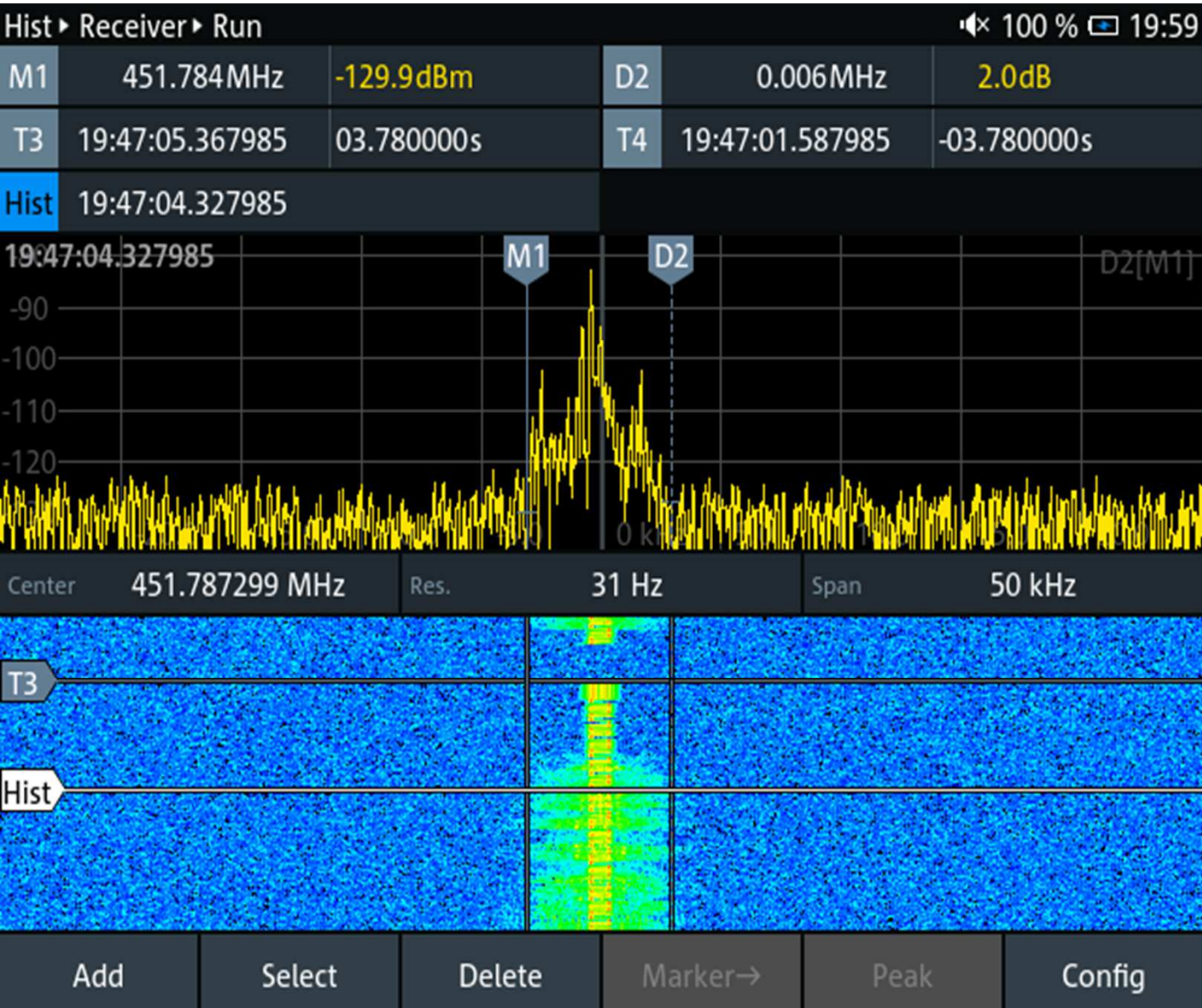
MARKERS AND LINES COMBINED

- ▶ M1/D2 show the signal is 6 kHz wide
- ▶ T3/T4 show transmission lasted for 3.76 secs



VIEWING HISTORY SPECTRUM

- ▶ HIST Marker allows user to “playback” Spectrum and Waterfall data
- ▶ HIST marker is moved to before signal became present



HISTORY SPECTRUM CONT.

- ▶ HIST marker now moved to where we can see the signal
- ▶ Enables user to gain understanding of signal characteristics/potential when signal of interest is no longer active

DISPLAY SETTINGS



ROHDE & SCHWARZ

Make ideas real



DISPLAY SETTINGS

- ▶ Display Settings can be accessed by using the “Disp” button
- ▶ This menu allows the user to customize the PR200 display:
 - Display Layout
 - Compass Location
 - Show/Hide Map Display
 - Scale
 - Waterfall Colors
- ▶ Note: Level Unit (dbm vs dbuv) can be found under Gear > User > Units and Formats or by pressing Gain > Config in any mode



DISPLAY LAYOUTS

- ▶ RX + Spectrum
- ▶ Spectrum
 - Amplitude vs Frequency
- ▶ Spectrum + Waterfall
- ▶ Waterfall
 - Time vs Amplitude
 - Red is strong, Green is medium, and Blue is weak
- ▶ Spectrum + Magnitude
 - Gated Spectrum
- ▶ HSCAN
- ▶ HSCAN + Spectrum



RX + SPECTRUM LAYOUT

*Connected
Antenna*

*Signal
Level*

*Level
Detector*

Receive frequency

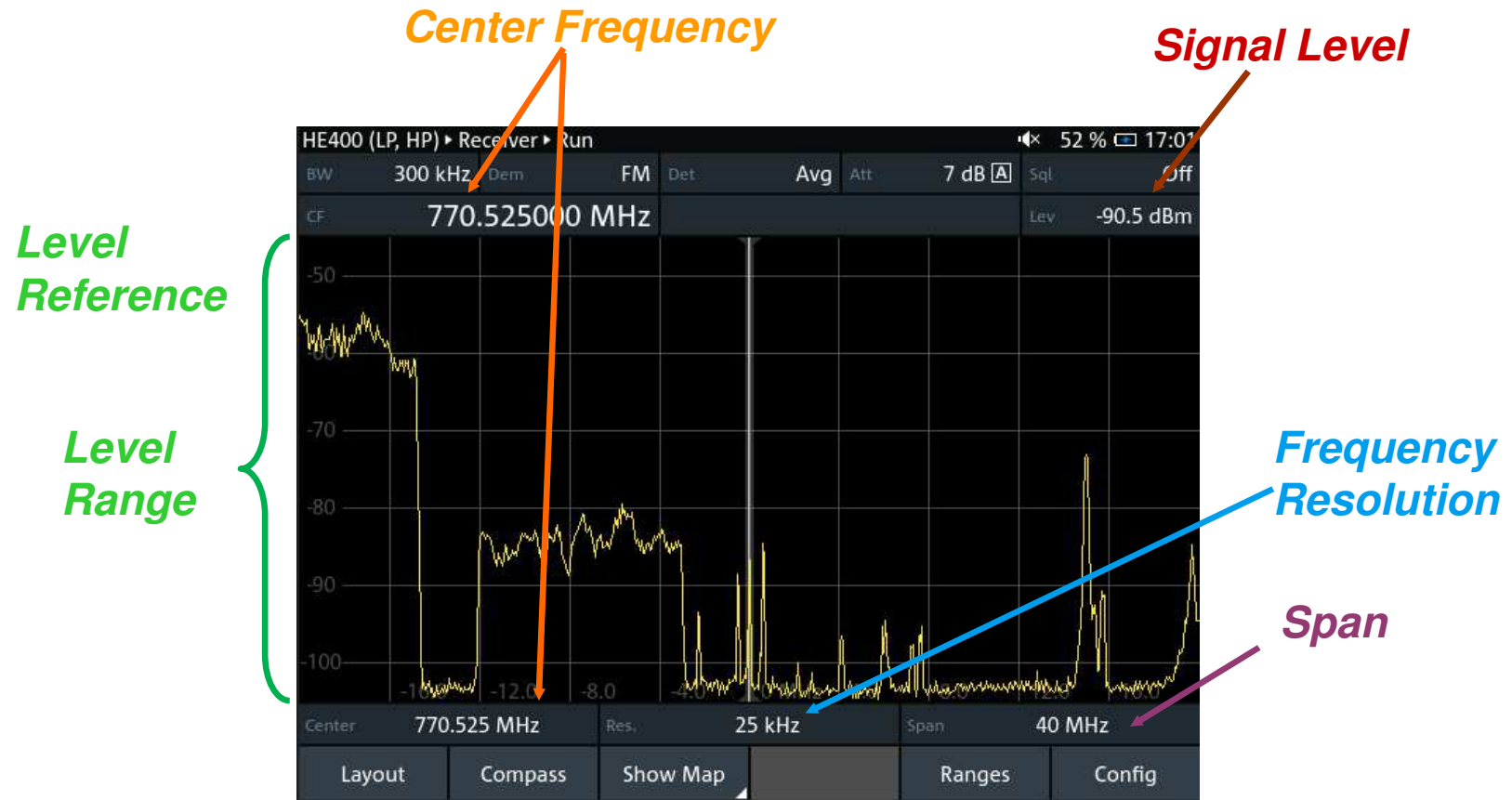
Frequency offset

Field strength

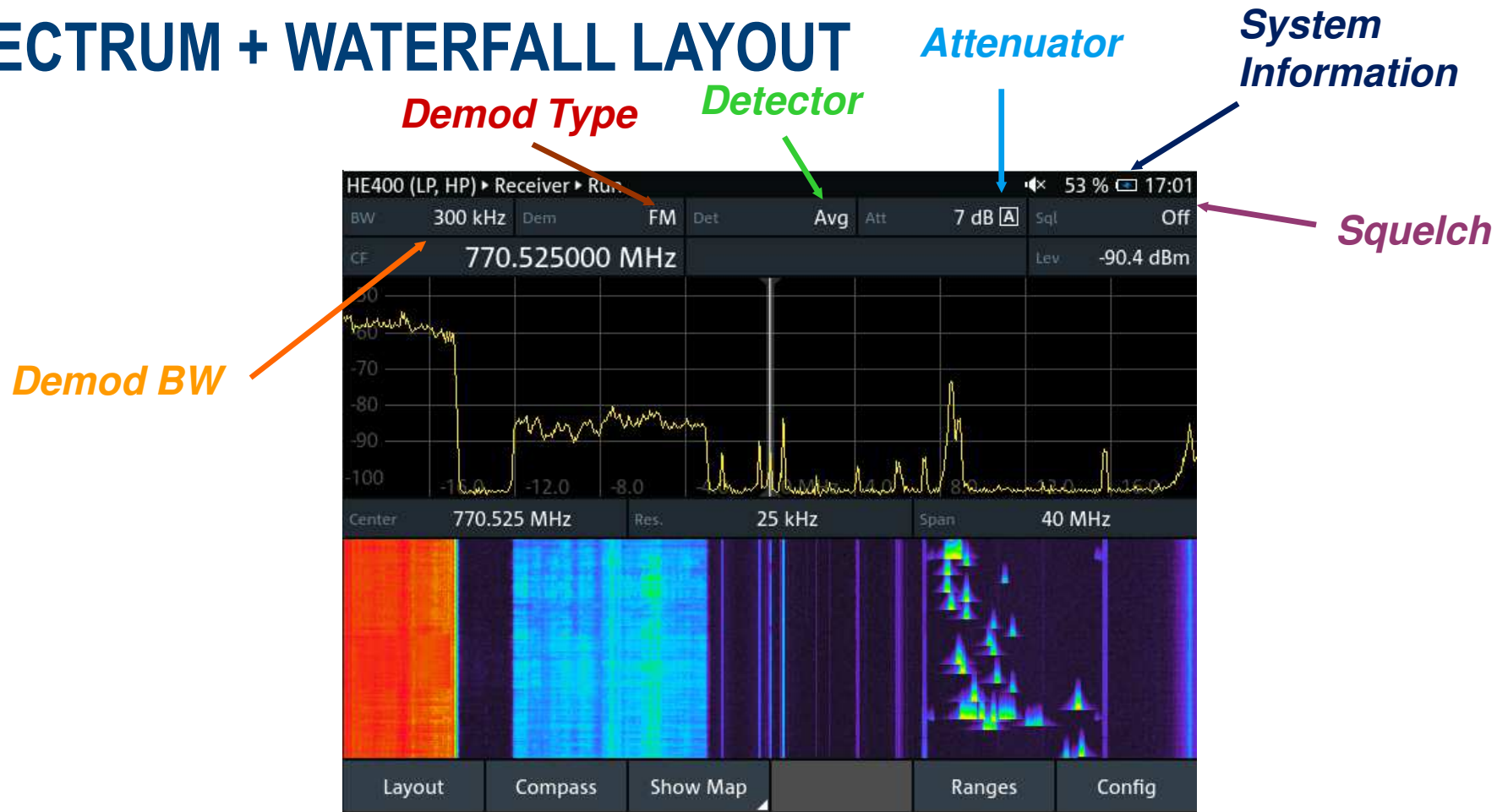
Span



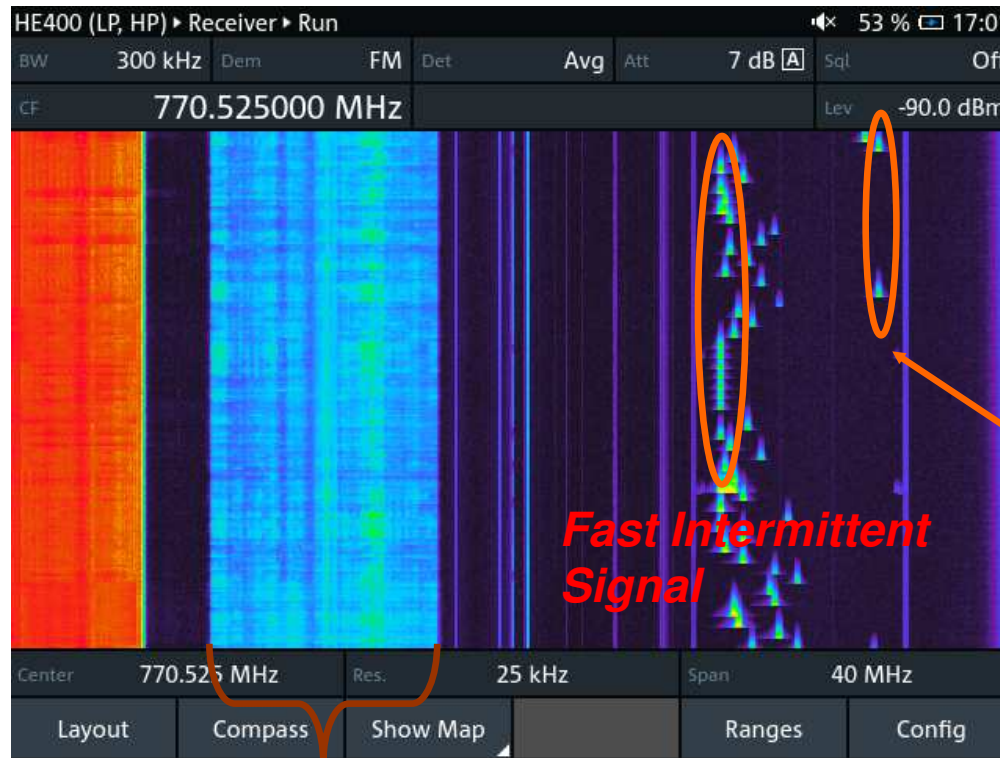
SPECTRUM LAYOUT



SPECTRUM + WATERFALL LAYOUT



WATERFALL LAYOUT

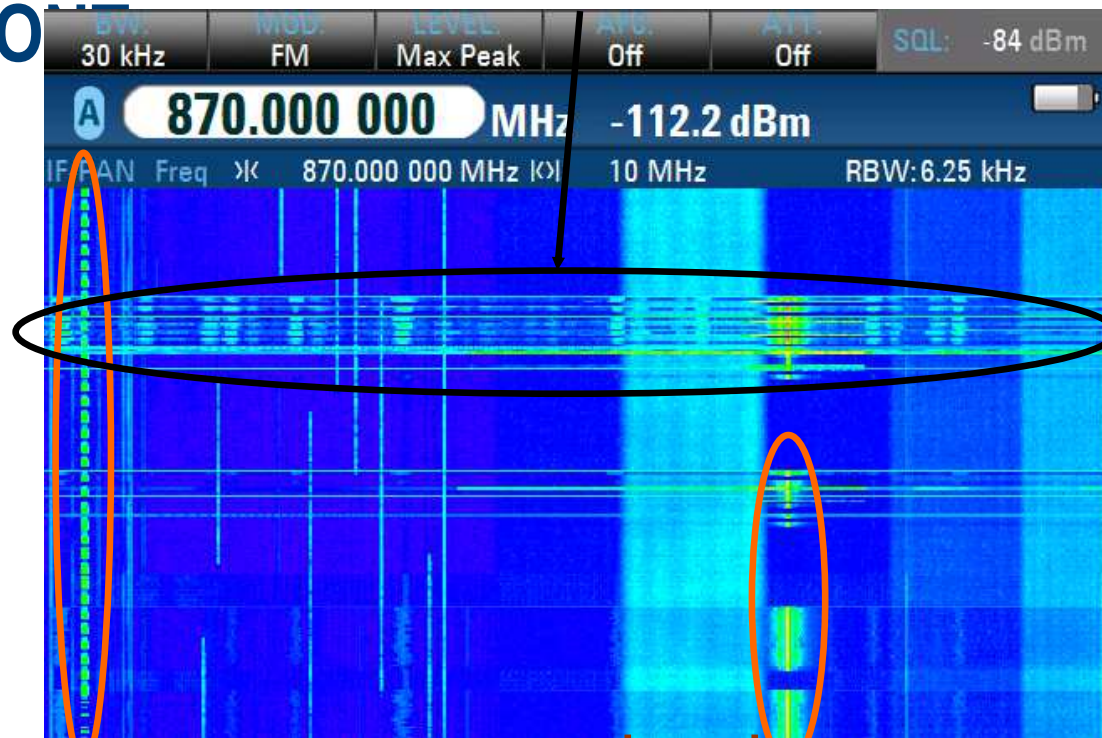


Constant carrier



WATERFALL CO

Broadband noise



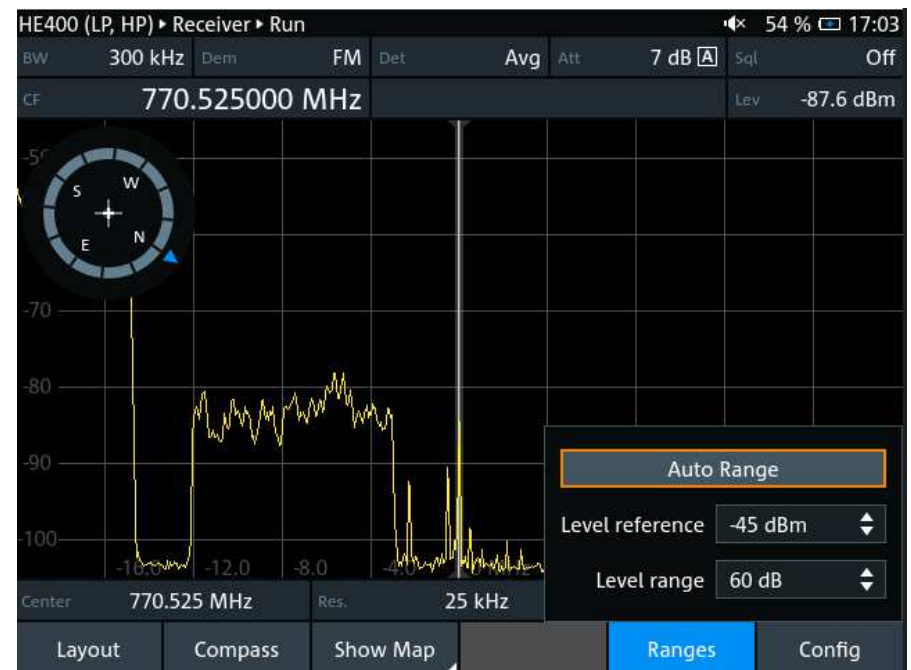
Fast intermittent signal

Constant carrier

Slow intermittent signal

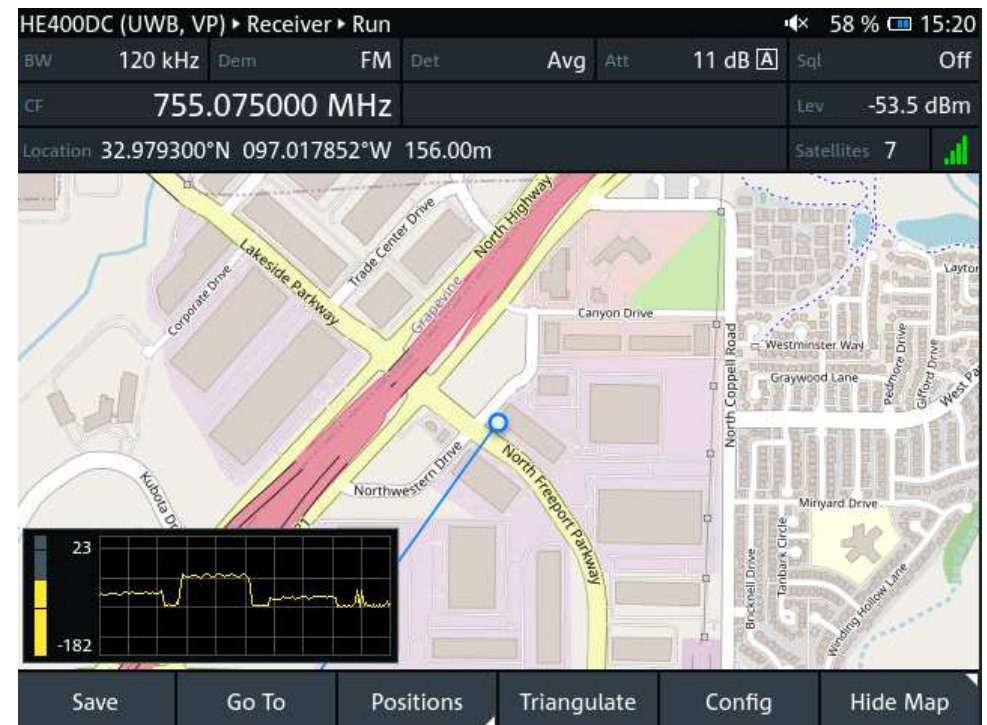
DISPLAY CONFIGURATION

- ▶ Compass – Allows the user to enable/disable a polar diagram showing which direction the directional antenna is pointed
 - This menu also allows you to change where the polar diagram is located on screen
- ▶ Range – Allows the user to set the spectrum Level Reference and Range manually or use Auto Range
 - Level Reference: Level at top of spectrum display
 - Level Range: How far down the spectrum display goes
- ▶ Config – Allows the user to configure water colors and squelch



MAP DISPLAY

- ▶ Shows current location and bearing azimuth
- ▶ Spectrum is overlaid on the map with level bar on left side
- ▶ Polar diagram will be overlaid on map as well (if in HSCAN mode)
- ▶ The map display can be used to manually locate a signal with saved bearings



THANK YOU

- ▶ Moving onto the next section:
- ▶ CISPR with PR200

